Semiparametric SDF Estimators for Pooled, Non-Traded Cash Flows

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Declaration of interest

The authors report no conflict of interest. The authors alone are responsible for the content and writing of the paper.

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Abstract

This paper introduces a unified semiparametric framework for estimating stochastic discount factors (SDF) on non-traded cash flow streams, providing a generalized objective-function perspective that nests the approach of Driessen et al. (2012) and conceptually relates to the time-series method of Korteweg and Nagel (2016). Our nonlinear least-mean-distance estimator is accompanied by a spatial asymptotic inference framework that accounts for the near-epoch dependence of overlapping private equity funds. Using Monte Carlo simulations, we demonstrate that the non-negligible finite-sample estimation error observed at short horizons can be controlled and effectively stabilized by averaging pricing errors over an extended set of compounding dates, and by aggregating cross-sectional data into seasoned vintage-year portfolios. Empirically, we analyze a comprehensive private equity dataset covering Buyout (BO) and Venture Capital (VC) funds and reveal two critical data constraints: First, including recent, unseasoned vintages severely inflates market beta estimates due to an outsized mechanical reliance on reported net asset values. When constraining the sample to heavily seasoned funds at an MSE-minimizing 15-year horizon, we estimate an attenuated market beta of 0.81 and 0.84 for equal- and value-weighted Buyout portfolios, respectively, and 1.30 and 1.07 for Venture Capital. Second, multi-factor models and abnormal performance (α) intercepts prove fundamentally fragile due to severe finite-sample noise. Consequently, we conclude that in our sample, single-factor market models are substantially more stable than multi-factor specifications, and we recommend restricting semiparametric SDF evaluation to such models until substantially more seasoned, cross-sectional private market data becomes available.

1 Introduction

Private equity has outgrown its niche, sitting today on more than $13 trillion in assets under management (McKinsey & Company, 2024), yet rigorous asset-pricing tools have not kept pace with this ascent. The empirical analysis and risk assessment of private equity and other non-traded cash flows remain fundamentally challenging due to the absence of market-based valuations and the inherent frictions of private markets (i.e., under incomplete information). Unlike public assets with trusted and tradeable valuations (in liquid secondary markets), private equity fund investments generate irregular, infrequently observed cash flows for which standard return-based asset pricing techniques are unsuitable.

To evaluate such irregular fund cash flows, the pioneering works of Driessen et al. (2012) and Korteweg and Nagel (2016) introduced highly influential stochastic discount factor (SDF) estimators that have become the gold standard in private market asset pricing. The profound impact of their methodologies is evident from their widespread adoption, as the Driessen et al. (2012) and Korteweg and Nagel (2016) estimators are heavily utilized to price a wide array of non-traded assets,

spanning private equity, venture capital and real estate to infrastructure and collateralized loan obligations. Building on these major contributions, we propose a unified semiparametric framework that recovers the Driessen et al. (2012) estimator as a special case and clarifies the mathematical connection to the Korteweg and Nagel (2016) approach. The core benefit of our unified framework lies in generalizing the set of discounting dates, which ultimately allows researchers to identify, comprehend, and control the small-sample bias inherent to cash-flow-based SDF pricing.

Our nonlinear least squares estimator stems from the class of Least-Mean-Distance (LMD) estimators, which are easier to handle than traditional Generalized Method of Moments (GMM) approaches (Pötscher and Prucha, 1997). Our LMD estimator both simplifies and extends the GMM methodology of Driessen et al. (2012), and we provide the spatial asymptotic inference framework that was missing in the original paper. The asymptotic inference formulations rely on the concept of spatial (near-epoch) dependence between funds as proposed by Korteweg and Nagel (2016). In this context, it is paramount to quantify the economic distance between funds by a measure like absolute vintage year difference or cash flow overlap[1].

For a typically sized dataset of private equity funds, we document a severe finite-sample bias in cash-flow-based SDF estimators like Driessen et al. (2012) and Korteweg and Nagel (2016) that inherently arises from the non-linearity of the estimating equations. While this bias can be mitigated by increasing the cross-sectional sample size, such data is rarely available. Instead, we demonstrate that by generalizing the estimator to average pricing errors over multiple, non-optimal discounting dates, researchers can deliberately introduce a "Timing Risk Premium" that effectively offsets this finite-sample bias, dramatically improving the estimates in small samples (in our simulation study).

In the empirical application of our new estimator, we evaluate simple linear SDF models utilizing the well-known Preqin cash flow database across a grid of compounding horizons. While evaluating multi-factor models proves unreasonable given high estimator variance, our single-factor market models yield stable and economically meaningful estimates for both Buyout and Venture Capital funds. At an MSE-minimizing 15-year horizon upon seasoned vintage years, we estimate a market beta of 0.81 and 0.84 for equal- and value-weighted Buyout portfolios, and 1.30 and 1.07 for Venture Capital, respectively (for a single-factor model without α intercept). While our rigorous asymptotic inference framework yields conservative standard errors, hv-block cross-validation stability diagnostics indicate that these market beta estimates are robust across data subsets. We document that the inherent estimator variance limiting multi-factor approaches can only be partially overcome by forming portfolios with a larger number of cross-sectional units per vintage year, highlighting a fundamental data constraint in modern private market asset pricing.

The paper is structured as follows. Section 2 reviews the related literature. Section 3 introduces our semiparametric LMD estimator and its corresponding asymptotic inference framework. Section 4 details the empirical implementation choices

[1]Note that this economic inter-fund *distance* is conceptually distinct from the statistical "distance" minimized by the Least-Mean-Distance estimator.

and validates the framework through a simulation study. Section 5 presents the empirical results for Buyout and Venture Capital funds, evaluating both single- and multi-factor SDF models. Section 6 concludes. Appendix A analyzes the bias–variance tradeoff in the context of our SDF estimation problem. Appendix B compares our simulation and empirical results to their Driessen et al. (2012) counterparts. Appendix C provides mathematical proofs.

2 Related literature

For private equity funds, performance evaluation under the SDF framework remains an active research topic (Driessen et al., 2012; Korteweg and Nagel, 2016, 2024; Giommetti and Jørgensen, 2021; Gupta and Nieuwerburgh, 2021; Boyer et al., 2023; Korteweg et al., 2024) alongside an expanding list of applications across other private asset classes. Specifically, the Driessen et al. (2012) estimator is heavily utilized to price human capital contracts, venture capital, real estate and private equity funds (Kroencke et al., 2013; Kim et al., 2015; Farrelly and Stevenson, 2019; Lee et al., 2019; Zhao, 2020; Tausch and Pietz, 2024), while the Korteweg and Nagel (2016) approach has found broad application spanning private equity, infrastructure, real estate investment trusts, collateralized loan obligations, impact funds and private debt (Giommetti and Jørgensen, 2021; Andonov et al., 2021; Riddiough and Wiley, 2022; Boyer et al., 2023; Cordell et al., 2023; Erel et al., 2024; Jeffers et al., 2024; Korteweg and Nagel, 2024). Previously, SDF performance evaluation was already applied for public stock markets by, e.g., Farnsworth et al. (2002) and for hedge fund returns by Li et al. (2016).

Within the private equity literature, SDF-based approaches can be broadly classified into parametric and semiparametric methods. Parametric models, such as those proposed by Franzoni et al. (2012) and Ang et al. (2018), explicitly specify the return dynamics and the underlying data generating process of private equity investments. While these methods can uncover latent factors or deal-level dynamics, they typically rely on strong distributional assumptions (e.g., log-normality) that may not fully capture the irregular and skewed nature of PE cash flows.

Consequently, semiparametric approaches have gained prominence, starting with the seminal cross-sectional GMM framework of Driessen et al. (2012). They estimate a linear factor model that minimizes the squared pricing errors of PE vintage year portfolios, explicitly evaluating discounted cash flows at fund inception. In a nutshell, the Driessen et al. (2012) estimator determines the estimated alpha and beta parameters of a linear factor model by minimizing the squared net present value (NPV) of all n fund cash flows in a least-squares optimization problem:

$$\left(\hat{\alpha}, \hat{\beta}\right) = \arg\min_{\alpha,\beta} \sum_{i=1}^{n} \left[NPV_i(\alpha, \beta)\right]^2 \tag{1}$$

with NPV for the ith fund

$$NPV_i(\alpha, \beta) = \sum_{t \in \{t_{0,i}, t_{0,i}+1, \dots\}} \frac{\text{CashFlow}_{i,t}}{\prod_{s=t_{0,i}+1}^{t} \left(1 + r_{f,s} + \alpha + \beta r_{\text{market},s}\right)}$$

Here, the Driessen et al. (2012) approach relies on the identifying assumption that the model is correctly specified, so that $\mathbb{E}[NPV_i(\theta_0)] = 0$. Building on this, Korteweg and Nagel (2016) introduced the Generalized Public Market Equivalent (GPME) framework, shifting the estimation of the SDF to public market data while evaluating PE cash flows in a second step. Importantly, Korteweg and Nagel (2016) were the first to apply a spatial heteroskedasticity and autocorrelation consistent (SHAC) covariance matrix estimator (Conley, 1999) to account for the economic distance (cash flow overlap) between PE funds, moving beyond simple cross-sectional bootstrapping.

Further complementing this cash flow dimension, a strand of literature models the informational frictions and smoothing inherent in reported Net Asset Values (NAVs) (Brown et al., 2019; Jenkinson et al., 2020; Brown et al., 2023). Additionally, Hüther et al. (2022) leverage a similar generalized framework to evaluate private firms using credit market equivalents. Finally, for comprehensive methodological and empirical overviews of risk and return assessment in private equity, we refer to Korteweg (2019, 2023), who also discuss the popular Dimson (1979) regression that employs lagged returns to correct for stale valuations.

Our paper's unified LMD framework contributes to this semiparametric literature by providing a loss-function formulation that encompasses the Driessen et al. (2012) approach and contextualizes the Korteweg and Nagel (2016) time-series analogue while supplying the rigorous increasing-domain asymptotic inference missing in earlier work (drawing on the spatial theory of Pötscher and Prucha (1997), Kim and Sun (2011), and Jenish and Prucha (2012)). While Gredil et al. (2020) highlighted that NPV-based measures are biased relative to per-period returns, we explicitly formalize this discrepancy as a "Timing Risk Premium", a concept linked to recent findings on discount-rate risk in secondary markets (Boyer et al., 2023).

3 Methodology

We now formalize a unified semiparametric framework for SDF estimation with pooled cash flows. Section 3.1 establishes the theoretical asset-pricing conditions and highlights how the choice of discounting date induces a Timing Risk Premium. Section 3.2 uses this insight to introduce a generalized Least-Mean-Distance (LMD) estimator that endogenizes the discounting dates. Section 3.3 builds an increasing-domain asymptotic framework to accommodate the spatial dependence of overlapping vintage years, and Section 3.4 describes large-sample inference. A detailed comparison demonstrating how this LMD estimator nests the framework of Driessen et al. (2012) as a special case and relates to Korteweg and Nagel (2016) is provided in Section 3.5.

3.1 Asset pricing for pooled cash flows

Because private equity funds invest capital over multiple quarters, any fund-level pricing restriction must aggregate deal-level moments across different investment dates. This subsection asks under what conditions such a pooled cash-flow stream satisfies a valid stochastic-discount-factor pricing restriction. The central finding is

Symbol	Description
n	Number of cross-sectional units (funds or portfolios)
V	Number of distinct vintage years
n_v	Number of portfolios in vintage year v
v_i	Vintage year of fund/portfolio i
t_i^0	Inception date of fund i
J_i	Number of deals for fund i
$K_{i,j}$	Number of divestment cash flows for deal j of fund i
$cf_{i,j,t}$	Net cash flow of deal j in fund i at time t
$CF_{i,t}$	Net cash flow of fund i at time t
$P_{\tau,t,i}$	Time-τ price of a single time-t cash flow
$PV_i(\theta)$	Pooled present value
$LCV_{i,\tau}$	Latent continuation value of fund i at date τ
$T, T_i^{\max}$	Final period and maximum fund lifetime
τ	Generic discounting or revaluation date
$\mathcal{F}_\tau$	Information set (filtration) at time τ
$\mathrm{Inv}_{i,j}, \mathrm{Div}_{i,j,k}$	Investment and divestment cash flows
$t_{i,j}^{\mathrm{Inv}}, t_{i,j,k}^{\mathrm{Div}}$	Deal investment and divestment dates
$m_{i,j}^{\mathrm{NFV}}(\cdot)$	NFV-type theoretical moment
Ψ_t , M_t	Multi-period and single-period SDF
$\Psi_{\tau,t}(\theta)$	SDF ratio Ψ_t/Ψ_τ, parameterized by θ
$\theta \in \mathbb{R}^p$	SDF parameter vector (e.g., intercept and factor loadings)
θ_0	True parameter vector
$\delta_{i,j}$	Deal-level pricing error

Table 1: Key notation for asset pricing and pooled cash flows (Section 3.1).

that the choice of discounting date is not innocuous: discounting at or before the first investment date preserves a clean conditional zero-moment condition (Lemma 1, Part 1), whereas post-investment discounting or future-value compounding generally introduces a timing-related covariance wedge (Parts 2 and 3). The argument proceeds in three steps. First, we state a deal-level decomposition of fund cash flows (Assumption 1) and derive the timing result at the deal level (Lemma 1). Second, we aggregate to the fund level and show that pooled Net Present Value (NPV) moments satisfy a valid zero conditional moment restriction whenever discounting occurs at or before fund inception (Proposition 1). Third, we discuss the implications for Net Future Value (NFV) moments and the interpretation of alpha measured ex post. Throughout, the deal-level structure serves only as a latent accounting device used to justify the fund-level restriction; only pooled fund cash flows are required as observable inputs to the estimator.

For fund i, let $CF_{i,t}$ denote the observed net cash flow at time t, and define the fund inception date by $t_i^0 := \min_j t_{i,j}^{\mathrm{Inv}}$. All fund-level variables are constructed from the following deal-level data generation process.

Assumption 1. ***Deal-level decomposition.*** *For each fund i, there exist J_i underlying deals such that:*

1. *Deal j has exactly one negative investment cash flow* $\mathrm{Inv}_{i,j} < 0$ *at time* $t_{i,j}^{\mathrm{Inv}} \in \{0, 1, \ldots, T-1\}$.
2. *Deal j has a finite stream of positive divestment cash flows* $(\mathrm{Div}_{i,j,k})_{k=1,\ldots,K_{i,j}}$ *with* $\mathrm{Div}_{i,j,k} > 0$ *and* $t_{i,j,k}^{\mathrm{Div}} > t_{i,j}^{\mathrm{Inv}}$ *for all* k.
3. *Let* $cf_{i,j,t}$ *denote the deal-level cash flow of deal j in fund i at time t. Then the observed fund-level cash flow is the cross-deal aggregate*

$$CF_{i,t} = \sum_{j=1}^{J_i} cf_{i,j,t} \qquad \forall i, t.$$

4. *The fund has no cash flows before inception:* $CF_{i,t} = 0$ *for all* $t < t_i^0$.

Under our central asset-pricing restriction, each deal is correctly priced by the stochastic discount factor at its investment date. Formally, the deal-level pricing error satisfies

$$\mathbb{E}\left[\delta_{i,j} \,\middle|\, \mathcal{F}_{t_{i,j}^{\mathrm{Inv}}}\right] = 0 \quad \forall\ i, j \tag{2}$$

where the deal-level pricing error is defined by

$$\delta_{i,j} := \mathrm{Inv}_{i,j} + \sum_{k=1}^{K_{i,j}} \frac{\Psi_{t_{i,j,k}^{\mathrm{Div}}}}{\Psi_{t_{i,j}^{\mathrm{Inv}}}} \mathrm{Div}_{i,j,k} \tag{3}$$

The next result characterizes how this deal-level pricing restriction depends on the choice of discounting or compounding date.

Lemma 1. *Suppose Equation* (2) *holds.*

1. *If* $\tau \le t_{i,j}^{\mathrm{Inv}}$, *then*

$$\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}\delta_{i,j} \,\middle|\, \mathcal{F}_\tau\right] = 0. \tag{4}$$

2. *If* $\tau > t_{i,j}^{\mathrm{Inv}}$, *then*

$$\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}\delta_{i,j} \,\middle|\, \mathcal{F}_\tau\right] = \frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}\,\mathbb{E}[\delta_{i,j} \mid \mathcal{F}_\tau], \tag{5}$$

which need not equal zero.

3. *Under the same pre-investment timing as Part (1), suppose additionally that the pricing error is compounded forward to a date* $t_{i,j}^{\mathrm{Future}} > t_{i,j}^{\mathrm{Inv}}$. *If* $\tau < t_{i,j}^{\mathrm{Inv}} < t_{i,j}^{\mathrm{Future}}$, *then define the NFV-type moment*

$$m_{i,j}^{\mathrm{NFV}}(\tau, t_{i,j}^{\mathrm{Future}}) := \frac{\Psi_{t_{i,j}^{\mathrm{Future}}}}{\Psi_\tau}\,\delta_{i,j}.$$

Its expected value

$$\mathbb{E}\left[m_{i,j}^{\mathrm{NFV}}(\tau, t_{i,j}^{\mathrm{Future}}) \mid \mathcal{F}_\tau\right] = \mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}\mathrm{Cov}\left(\frac{\Psi_{t_{i,j}^{\mathrm{Future}}}}{\Psi_{t_{i,j}^{\mathrm{Inv}}}}, \delta_{i,j} \mid \mathcal{F}_{t_{i,j}^{\mathrm{Inv}}}\right) \mid \mathcal{F}_\tau\right] \tag{6}$$

generally contains a conditional covariance term and therefore need not equal zero.

Proof. Consider part (1). Applying the law of iterated expectations,[2] since $\tau \leq t_{i,j}^{\text{Inv}}$, the SDF ratio is known at $t_{i,j}^{\text{Inv}}$ and can be factored out yielding

$$\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\delta_{i,j} \mid \mathcal{F}_\tau\right] = \mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau} \underbrace{\mathbb{E}\left[\delta_{i,j} \mid \mathcal{F}_{t_{i,j}^{\text{Inv}}}\right]}_{=0 \text{ by Equation (2)}} \mid \mathcal{F}_\tau\right] = 0.$$

For part (2), since $t_{i,j}^{\text{Inv}} < \tau$, the ratio $\Psi_{t_{i,j}^{\text{Inv}}}/\Psi_\tau$ is $\mathcal{F}_\tau$-measurable, so

$$\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\delta_{i,j} \,\middle|\, \mathcal{F}_\tau\right] = \frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\mathbb{E}[\delta_{i,j} \mid \mathcal{F}_\tau].$$

Since some divestment cash flows of deal j may have been paid by date τ, $\delta_{i,j}$ is partially $\mathcal{F}_\tau$-measurable, and its conditional expectation given $\mathcal{F}_\tau$ generally reflects realized outcomes rather than the zero-mean restriction at $t_{i,j}^{\text{Inv}}$.

For part (3), using the law of iterated expectations with an intermediate conditioning on $\mathcal{F}_{t_{i,j}^{\text{Inv}}}$,

$$\begin{aligned}\mathbb{E}\left[m_{i,j}^{\text{NFV}}(\tau, t_{i,j}^{\text{Future}}) \mid \mathcal{F}_\tau\right] &= \mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Future}}}}{\Psi_{t_{i,j}^{\text{Inv}}}}\delta_{i,j} \mid \mathcal{F}_{t_{i,j}^{\text{Inv}}}\right] \mid \mathcal{F}_\tau\right] \\ &= \mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\text{Cov}\left(\frac{\Psi_{t_{i,j}^{\text{Future}}}}{\Psi_{t_{i,j}^{\text{Inv}}}}, \delta_{i,j} \mid \mathcal{F}_{t_{i,j}^{\text{Inv}}}\right) \mid \mathcal{F}_\tau\right]\end{aligned}$$

where we used the conditional covariance identity and Equation (2). □

Part (3) shows that compounding deal cash flows to a future date generally destroys the zero conditional moment restriction because the resulting expression contains a covariance between the pricing error and future SDF realizations. We refer to this non-zero expected value as the *timing wedge*. Economically, this wedge represents a **Timing Risk Premium**, i.e., the required compensation for the discount-rate risk borne during the interim illiquidity period between investment and realization. Accordingly, NFV-type moments are inappropriate as primitive ex-ante pricing moments unless this timing term is modeled explicitly. They can still be informative for ex-post benchmarking, but the resulting performance measure must then be interpreted as a composite object that mixes true abnormal performance (skill) with timing-related compensation.

Proposition 1. ***Price of a pooled cash-flow stream at or before fund inception.*** *If $\tau \leq t_i^0$, then*

$$\mathbb{E}\left[\sum_{t=\tau}^{T}\frac{\Psi_t}{\Psi_\tau}CF_{i,t} \,\middle|\, \mathcal{F}_\tau\right] = 0. \tag{7}$$

[2] At the boundary $\tau = t_{i,j}^{\text{Inv}}$, the SDF ratio $\Psi_{t_{i,j}^{\text{Inv}}}/\Psi_\tau = 1$ is a trivial constant, so the covariance with $\delta_{i,j}$ is zero and the same argument applies.

Proof. If $\tau \leq t_i^0$, then $\tau \leq t_{i,j}^{\text{Inv}}$ for every deal j in fund i. Using the deal-level decomposition of the pooled cash-flow stream and Equation (3),

$$\mathbb{E}\left[\sum_{t=\tau}^{T} \frac{\Psi_t}{\Psi_\tau} CF_{i,t} \,\middle|\, \mathcal{F}_\tau\right] = \mathbb{E}\left[\sum_{j=1}^{J_i} \frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau} \delta_{i,j} \,\middle|\, \mathcal{F}_\tau\right]. \tag{8}$$

Applying Equation (4) term-by-term yields

$$\mathbb{E}\left[\sum_{j=1}^{J_i} \frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau} \delta_{i,j} \,\middle|\, \mathcal{F}_\tau\right] = 0. \tag{9}$$

□

Proposition 1 provides the clean benchmark for identification: pooled NPV moments are theoretically well founded when discounting occurs at or before fund inception. Once $\tau > t_i^0$, some underlying deals are active or already partially realized, so the corresponding pooled pricing error incorporates the timing wedge. The choice of discounting dates is therefore not innocuous. Section 3.2 formalizes this choice by defining an estimator that explicitly specifies which discounting dates are used: when it restricts to dates $\tau \leq t_i^0$, the pooled fund-level pricing error inherits a valid zero conditional moment; when it includes dates $\tau > t_i^0$, it generally targets a tilted criterion that embeds the Timing Risk Premium characterized by Equation (6).

Critically, this Timing Risk Premium is not solely relevant for ex-post benchmarking. While it dictates that any *alpha* measured via Net Future Values is mechanically a composite metric (as detailed in Appendix A.6), it also fundamentally drives the large-sample and small-sample properties of our generalized LMD estimator. As we demonstrate in later sections, averaging pricing errors over extended horizons deliberately injects this structural timing wedge to effectively counterbalance severe finite-sample estimator bias and variance.

3.2 Least-Mean-Distance estimator

In this subsection, we define a generalized fund-level criterion that averages pricing errors across a user-specified set of discounting dates $\mathcal{T}_i$. This nests the inception-only NPV criterion of Driessen et al. (2012) as a special case and allows us to study how horizon averaging trades structural validity against finite-sample regularization. Throughout, only pooled fund cash flows are observed, whereas the deal-level decomposition from Section 3.1 remains a latent theoretical device.

The stochastic discount factor $\Psi_{\tau,t}$ is used to calculate the time-τ price $P_{\tau,t,i}$ of a **single** time-t cash flow of any given PE fund i

$$P_{\tau,t,i} := \Psi_{\tau,t} \cdot CF_{i,t} = \frac{\Psi_t}{\Psi_\tau} \cdot CF_{i,t} \qquad \forall \quad \tau, t, i \tag{10}$$

with multi-period SDF $\Psi_t = \prod_{k=1}^{t} M_k$ and $\Psi_0 = 1$. As SDFs are commonly parameterized by a vector $\theta \in \mathbb{R}^p$, i.e., $\Psi_{\tau,t} \equiv \Psi_{\tau,t}(\theta)$, our goal is to find an estimation method for the true parameter vector θ_0.

Symbol	Description
$\epsilon_{\tau,i}$	Fund pricing error at discounting date τ
$\bar{\epsilon}_i, \bar{\epsilon}_i^{(H)}$	Weighted average pricing error (generic and horizon-specific)
H	Horizon parameter
$\mathcal{T}_i, \mathcal{T}_{i,H}$	Set of discounting dates (generic and horizon-specific)
w_i	Weighting factor for fund i
$L(\cdot)$	Loss function (e.g., $L(x) = x^2$)
$Q_H(\theta)$	Horizon-specific population objective function
$Q_n(\theta), Q_{n,H}(\theta)$	Sample objective function (generic and horizon-specific)
$s_i^{(H)}(\theta)$	Unit score contribution
$d_{i,j}$	Economic distance between units i and j
$K(\cdot), k_{i,j}$	Symmetric SHAC kernel function and generic weights
b_V	SHAC bandwidth parameter
$\hat{\theta}, \hat{\theta}_H$	LMD Estimator (generic and horizon-specific targets)
$\theta_H^\dagger$	Horizon-specific pseudo-true parameter vector
$\mathcal{H}_H$	Hessian of population objective at $\theta_H^\dagger$
Λ_H	Long-run covariance matrix of the score
Σ_H	Asymptotic covariance matrix of $\hat{\theta}_H$
W_H	Wald statistic for linear hypotheses
R, r	Linear constraint matrix and vector

Table 2: Key notation for the LMD estimator and asymptotic inference (Sections 3.2–3.4).

For each fund i and all points τ within a common fund lifetime, the empirical pricing error $\epsilon_{\tau,i}(\theta)$ of **all** fully realized fund cash flows is calculated as a time-τ revaluation of the entire cash-flow stream

$$\epsilon_{\tau,i}(\theta) := \sum_{t=0}^{T} P_{\tau,t,i}(\theta) \qquad \forall \quad \tau, i \tag{11}$$

Note that for $t < \tau$, the terms $\frac{\Psi_t}{\Psi_\tau} CF_{i,t}$ are compounded past cash flows, not discounted future cash flows, so the term "present value" is no longer accurate for $\tau > t_i^0$. For $\tau > t_i^0$, $\epsilon_{\tau,i}(\theta)$ is not a continuation value but a time-τ revaluation of the entire realized cash-flow stream, and therefore generally inherits the timing wedge characterized in Section 3.1. This time-τ revaluation $\epsilon_{\tau,i}(\theta)$ must be sharply distinguished from the latent continuation value $LCV_{i,\tau}(\theta) := \mathbb{E}[\sum_{t=\tau}^{T}(\Psi_t/\Psi_\tau)CF_{i,t} \mid \mathcal{F}_\tau]$, which conditions on information at τ and is not directly observed.

To analyze the effect of different discounting dates τ on the bias and variance of the estimated SDF parameters, we define the generalized criterion $\bar{\epsilon}_i(\theta)$ that averages pricing errors over the set $\mathcal{T}_i$:

$$\bar{\epsilon}_i(\theta) = w_i \cdot \frac{1}{\text{card}(\mathcal{T}_i)} \sum_{\tau \in \mathcal{T}_i} \epsilon_{\tau,i}(\theta) \qquad \forall \quad i \tag{12}$$

where w_i normalizes fund-level pricing errors before the loss is applied; it is not

an observation weight in the outer sample average. Depending on the application, w_i may normalize by invested capital, by aggregate vintage-year capital, by a macroeconomic deflator, or be set to one when cash flows are already expressed in comparable scale units.

The role of $\mathcal{T}_i$ The set $\mathcal{T}_i$ controls which discounting dates enter the averaged pricing error $\bar{\epsilon}_i(\theta)$ in Equation (12) (see Figure 1). Averaging over multiple dates in $\mathcal{T}_i$ was originally motivated by the "exploding alpha" problem first described by Driessen et al. (2012): when the initial cash flow is very small, an arbitrarily large alpha can discount all subsequent cash flows close to zero, rendering the beta factors irrelevant and distorting estimation. Averaging over $\mathcal{T}_i$ mitigates this degeneracy. The relevant horizon family is parameterized as

$$\mathcal{T}_{i,H} = \left\{t_i^0, t_i^0 + 1, \ldots, \min(t_i^0 + 12H, T_i^{\max})\right\}, \qquad H \geq 0. \tag{13}$$

This formulation encompasses two economically distinct cases:

- $H = 0$: If $\mathcal{T}_i = \{t_i^0\}$, the estimator uses an inception-only NPV criterion, is anchored in the pooled pricing restriction from Section 3.1, and targets the structural parameter θ_0.
- $H > 0$: If $\mathcal{T}_i = \mathcal{T}_{i,H}$ with $H > 0$, the estimator is a generalized horizon-averaged criterion and targets the pseudo-true parameter $\theta_H^\dagger$. It provides finite-sample regularization benefits but incorporates a potential timing distortion.

Larger values of the horizon H offer more regularization, but a larger departure from the structural benchmark with zero expected value.

Remark 1 (Target parameter under horizon averaging)**.** *For $\mathcal{T}_{i,H} = \{t_i^0, t_i^0 + 1, \ldots, \min(t_i^0 + 12H, T_i^{\max})\}$, the estimator targets the horizon-specific pseudo-true parameter $\theta_H^\dagger = \arg\max_\theta Q_H(\theta)$. This coincides with the structural parameter θ_0 only when $H = 0$. For $H > 0$, the estimator trades structural fidelity against finite-sample regularization by averaging pricing errors over discounting dates that may incorporate the timing wedge from Section 3.1.*

Restriction used for identification and asymptotic theory. In the consistency analysis below, we specialize the generalized criterion to $\mathcal{T}_i = \{t_i^0\}$. Under Proposition 1, this yields a valid pooled fund-level zero-moment restriction at the structural parameter θ_0.

Remark 2. *This restriction guarantees that $\mathbb{E}[\bar{\epsilon}_i(\theta_0)] = 0$ under the true parameter. For the quadratic loss $L(x) = x^2$, the population objective $Q_0(\theta) = -\mathbb{E}[\bar{\epsilon}_i(\theta)^2] = -\mathrm{Var}[\bar{\epsilon}_i(\theta)] - (\mathbb{E}[\bar{\epsilon}_i(\theta)])^2$ involves both variance and bias terms. Consistency then requires that θ_0 uniquely minimizes this composite criterion (see Proposition 3).*

Because our estimator is defined through the optimization of a sample criterion, it is an extremum estimator in the sense of Newey and McFadden (1994, Eq. 1.1), with $\hat{\theta} = \arg\max_{\theta \in \Theta} Q_n(\theta)$.

Concretely, our Least Mean Distance (LMD) estimator (Pötscher and Prucha, 1997, Equation 7.1) minimizes the average loss of $\bar{\epsilon}_i(\theta)$. For consistency with the

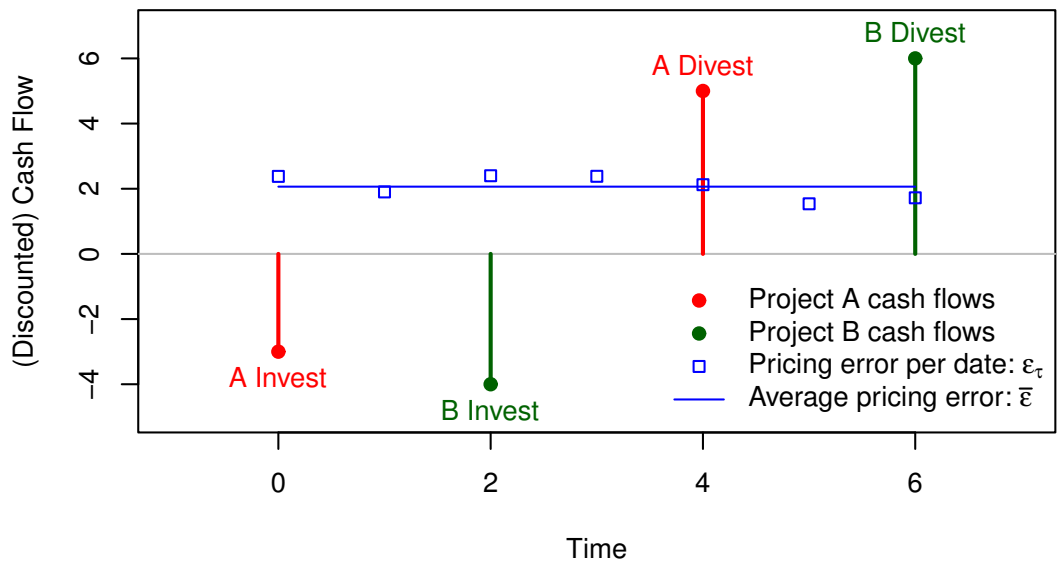

Figure 1: How to calculate and interpret the average pricing error introduced by Equation (12): The time index t is relevant for the net cash flows (black dots). The time index τ is used for the pricing error, i.e., the sum of discounted net cash flows (blue boxes). The weighted average of these "pricing errors" gives the average pricing error $\bar{\epsilon}$ as defined in Equation (12) (solid blue line). In this example, $\text{card}(\mathcal{T}_i) = 7$, i.e., the number of blue boxes.

extremum-estimator convention, we will technically maximize the average negative loss of $\bar{\epsilon}_i(\theta)$, which is equivalent.

$$\hat{\theta} = \arg\min_{\theta\in\Theta} \ \frac{1}{n}\sum_{i=1}^{n} L\big(\bar{\epsilon}_i(\theta)\big) \equiv \arg\max_{\theta\in\Theta} \ Q_n(\theta) \tag{14}$$

with

$$Q_n(\theta) = -\frac{1}{n}\sum_{i=1}^{n} L\big(\bar{\epsilon}_i(\theta)\big)$$

where $L : \mathbb{R}^1 \to \mathbb{R}^1$ denotes a loss function, e.g., $L(x) = (x-0)^2$ in the case of nonlinear least squares. Throughout the paper, the weighted average fund pricing error $\bar{\epsilon}_i \equiv \bar{\epsilon}_i(\theta)$ is regarded as a nonlinear random function of the SDF parameter θ.

3.3 Asymptotic framework

The large-sample analysis is stated for the horizon-indexed estimator

$$\hat{\theta}_H = \arg\max_{\theta\in\Theta} Q_{n,H}(\theta), \qquad Q_{n,H}(\theta) := -\frac{1}{n}\sum_{i=1}^{n} L\Big(\bar{\epsilon}_i^{(H)}(\theta)\Big), \tag{15}$$

where

$$\bar{\epsilon}_i^{(H)}(\theta) := w_i \cdot \frac{1}{\text{card}(\mathcal{T}_{i,H})} \sum_{\tau\in\mathcal{T}_{i,H}} \epsilon_{\tau,i}(\theta)$$

and

$$\mathcal{T}_{i,H} = \left\{ t_i^0, t_i^0 + 1, \ldots, \min(t_i^0 + 12H, T_i^{\max}) \right\}.$$

Throughout this subsection, the horizon parameter $H \geq 0$ is fixed while $n \to \infty$.

The notation in Equation (15) makes explicit that the estimator and its probability limit generally depend on the chosen horizon H. This distinction is essential in the present setting because the pooled inception-based pricing restriction yields a structural zero-moment condition only for $H = 0$, whereas horizon averaging with $H > 0$ may introduce the timing wedge discussed in Section 3.1.

Population criterion and target parameter. Define the horizon-specific population objective by

$$Q_H(\theta) := \mathbb{E}[Q_{n,H}(\theta)] = -\mathbb{E}\Big[L\Big(\bar{\epsilon}_i^{(H)}(\theta)\Big)\Big]. \tag{16}$$

Under the ergodicity implied by the increasing-domain framework (Assumption 3), the Cesàro average $n^{-1}\sum_{i=1}^{n} \mathbb{E}[L(\bar{\epsilon}_i^{(H)}(\theta))]$ converges to the single-unit expectation as $V \to \infty$. The corresponding population target is the unique maximizer

$$\theta_H^\dagger := \arg\max_{\theta \in \Theta} Q_H(\theta). \tag{17}$$

Remark 3 (Structural versus horizon-specific target)**.** *The asymptotic theory below is intentionally formulated around $\theta_H^\dagger$, not around θ_0. This distinction is substantive rather than cosmetic. For $H = 0$, the pooled inception-only criterion coincides with the structural pricing restriction of Section 3.1, and hence $\theta_0^\dagger = \theta_0$. For $H > 0$, the criterion generally targets a pseudo-true parameter $\theta_H^\dagger \neq \theta_0$, because averaging over $\tau > t_i^0$ may embed the timing wedge at the population level.*

This formulation keeps the asymptotic statements aligned with the estimator actually implemented in the empirical analysis. It also cleanly separates two conceptually distinct questions: (i) the large-sample behavior of the estimator relative to its own criterion, and (ii) the economic interpretation of that criterion relative to the structural parameter θ_0.

3.3.1 Vintage-year increasing-domain asymptotics

By the value-additivity principle (Hansen and Richard, 1987), the SDF restriction can be formulated for pooled cash flows at different aggregation levels. In the empirical implementation, we form vintage-year portfolios as cross-sectional units, because this reduces idiosyncratic fund-level noise while preserving economically meaningful variation across macro environments.

Assumption 2. *For each vintage year v, we pool fund cash flows to form n_v portfolios, so that*

$$n = \sum_{v=1}^{V} n_v.$$

The boundary cases are: (i) single-fund portfolios and (ii) one portfolio per vintage year. Both equal-weighted and value-weighted portfolio constructions are admissible.

Without loss of generality, we continue to refer to the resulting cross-sectional units as funds. Let $v_i := \mathrm{YEAR}(t_i^0)$ denote the vintage year of unit i.

The pricing error $\bar{\epsilon}_i^{(H)}(\theta)$ is indexed by both unit identity and vintage year. Accordingly, the data are naturally represented as an increasing-domain random field rather than as a standard iid cross-section or a conventional long time series. Economically, dependence arises because adjacent vintage years are exposed to overlapping macroeconomic conditions and partially overlapping realized cash-flow windows.

Assumption 3 (Vintage-year increasing-domain asymptotics)**.** *As $n \to \infty$:*

1. *The number of distinct vintage years satisfies $V \to \infty$.*
2. *The number of units per vintage year is uniformly bounded: there exists $\bar{n} < \infty$ such that $n_v \leq \bar{n}$ for all v.*
3. *Fund lifetimes are uniformly bounded: there exists $\bar{T} < \infty$ such that $T_i^{\max} - t_i^0 \leq \bar{T}$ for all i.*
4. *The economic distance between units i and j is measured by*

$$d_{i,j} = |v_i - v_j| + \rho_0 \mathbf{1}_{\{i \neq j\}}, \qquad \rho_0 > 0.$$

Assumption 3 imposes increasing-domain asymptotics and rules out infill asymptotics in the sense of Jenish and Prucha (2012). The role of $\rho_0 > 0$ is to prevent the field from becoming asymptotically dense at a fixed location. This is economically natural here: asymptotic information is generated by observing additional vintage years, not by observing an ever denser sample from a fixed macroeconomic environment.

A useful implication of Assumption 3(2) is that the total number of observations n and the number of vintages V are of the same order. Indeed,

$$V \leq n = \sum_{v=1}^{V} n_v \leq \bar{n} V,$$

so $n \asymp V$. Consequently, $\sqrt{n}$- and $\sqrt{V}$-rates are asymptotically equivalent up to a constant factor. Put differently, statistical information grows with the span of vintage years rather than with within-vintage clustering.

This increasing-domain perspective differs from the infill framework of Driessen et al. (2012), who fix the number of portfolios and let the number of underlying funds per portfolio diverge. Their asymptotic device is appropriate for consistency conditional on a fixed macroeconomic environment. By contrast, the present framework is designed for unconditional identification and inference when new vintages generate new realizations of macroeconomic conditions.

3.3.2 Uniform convergence of the criterion

The first step is a uniform law of large numbers for the horizon-specific sample criterion $Q_{n,H}(\theta)$.

Assumption 4 (Parameter space)**.** *The parameter space $\Theta \subset \mathbb{R}^p$ is compact.*

Assumption 5 (Primitive regularity conditions for uniform convergence)**.** *For each fixed H:*

1. *The array*
$$\left\{\bar{\epsilon}_i^{(H)}(\theta) : i = 1, \ldots, n\right\}$$
is near-epoch dependent (NED) on an α-mixing vintage-year process $\{\eta_v\}_{v\geq 1}$, uniformly in $\theta \in \Theta$ (that is, the NED approximation constants are bounded uniformly over $\theta \in \Theta$ and i).

2. *The loss function $L(\cdot)$ is continuous and Lipschitz on the relevant support.*

3. *There exists $p > 2$ such that*
$$\sup_{\theta\in\Theta} \mathbb{E}\left[\left|L\left(\bar{\epsilon}_i^{(H)}(\theta)\right)\right|^p\right] < \infty$$
uniformly in i.

4. *The criterion is stochastically equicontinuous in θ.*

Assumption 5 is stated at a deliberately primitive level: dependence is imposed on the underlying horizon-specific pricing errors, and the loss inherits the required regularity through continuity and Lipschitz continuity. This is preferable to assuming a law of large numbers directly for the transformed criterion because it ties the statistical assumptions more closely to the economic structure of the data.

Proposition 2 (Uniform law of large numbers)**.** *Under Assumptions 2, 3, and 5,*
$$\sup_{\theta\in\Theta} |Q_{n,H}(\theta) - Q_H(\theta)| \xrightarrow{p} 0.$$

Outline of Proof. The result follows from a uniform law of large numbers for NED random fields on α-mixing bases; see, for example, Jenish and Prucha (2009, Theorem 6). The bounded-lifetime and bounded-within-vintage assumptions ensure that cross-sectional overlap remains local in vintage-year distance, while the moment and equicontinuity conditions deliver uniform stochastic control over $\theta \in \Theta$. □

Economically, Proposition 2 says that when the sample spans sufficiently many distinct vintages, the empirical criterion stabilizes around its population counterpart. This remains true whether the horizon is structural ($H = 0$) or regularized ($H > 0$); what changes across horizons is the population target itself.

3.3.3 Consistency

Assumption 6 (Identification of the horizon-specific target)**.** *For each fixed H, the population criterion $Q_H(\theta)$ has a unique maximizer*
$$\theta_H^\dagger = \arg\max_{\theta\in\Theta} Q_H(\theta).$$

Proposition 3 (Consistency)**.** *Under Assumptions 2, 3, 4, 5, and 6,*
$$\hat{\theta}_H \xrightarrow{p} \theta_H^\dagger \qquad \text{as } n \to \infty.$$

Outline of Proof. The claim follows from the standard extremum-estimator argument of Newey and McFadden (1994, Theorem 2.1). Proposition 2 yields uniform convergence of $Q_{n,H}$ to Q_H, Assumption 6 provides a unique maximizer, and compactness of Θ (Assumption 4) together with continuity of the discount-factor mapping $\Psi_{\tau,t}(\theta)$ and the loss function deliver the remaining conditions. □

Corollary 1 (Structural consistency under inception-only pricing)**.** *Suppose, in addition, that $H = 0$ and that the pooled inception-only pricing restriction of Proposition 1 holds at the structural parameter θ_0. Then*

$$\theta_0^\dagger = \theta_0 \qquad \textit{and hence} \qquad \hat{\theta}_0 \xrightarrow{p} \theta_0.$$

Corollary 1 isolates the special case in which the estimator converges to the structural SDF parameter. This separation is useful empirically because it prevents the reader from conflating consistency of the extremum estimator with structural validity of a horizon-averaged criterion.

3.3.4 Central limit theorem

Building on the consistency result in Proposition 3, we now establish the asymptotic distribution of the estimator. Like consistency, the central limit theorem is formulated around the horizon-specific target $\theta_H^\dagger$. This requires a standard local-convergence condition on the sample Hessian.

Assumption 7 (Local Hessian convergence)**.** *For each fixed H, there exists an open neighborhood $\mathcal{N}_H$ of $\theta_H^\dagger$ such that*

$$\sup_{\theta \in \mathcal{N}_H} \|\nabla_{\theta\theta} Q_{n,H}(\theta) - \nabla_{\theta\theta} Q_H(\theta)\| \xrightarrow{p} 0.$$

Assumption 7 is the standard sample-Hessian convergence condition used in extremum-estimator central limit theorems (e.g., Newey and McFadden, 1994). In our increasing-domain framework, it follows from the same random-field LLN argument as Proposition 2, provided that the Hessian contributions inherit the required NED, moment, and stochastic-equicontinuity properties from the pricing errors $\bar{\epsilon}_i^{(H)}(\theta)$ and the loss function $L(\cdot)$.

Proposition 4 (Asymptotic normality around the horizon-specific target)**.** *Under the conditions of Proposition 3 and Assumption 7, suppose additionally that:*

1. *$\theta_H^\dagger$ lies in the interior of Θ;*
2. *$Q_H(\theta)$ is twice continuously differentiable in a neighborhood of $\theta_H^\dagger$;*
3. *the Hessian*
$$\mathcal{H}_H := \nabla_{\theta\theta} Q_H(\theta_H^\dagger)$$
is nonsingular; and
4. *a central limit theorem holds for the score contributions*
$$s_i^{(H)}(\theta) := \nabla_\theta L\left(\bar{\epsilon}_i^{(H)}(\theta)\right).$$

Then

$$\sqrt{n}\left(\hat{\theta}_H - \theta_H^\dagger\right) \xrightarrow{d} \mathcal{N}(0, \Sigma_H),$$

where

$$\Sigma_H = \mathcal{H}_H^{-1}\Lambda_H\mathcal{H}_H^{-1}, \qquad \Lambda_H := \lim_{n\to\infty} n \cdot \mathrm{Var}\left[\nabla_\theta Q_{n,H}(\theta_H^\dagger)\right].$$

Outline of Proof. The result follows from the extremum-estimator CLT of Newey and McFadden (1994, Theorem 3.1) combined with a central limit theorem for NED random fields under increasing-domain asymptotics; see Jenish and Prucha (2012, Theorem 4). The nonstandard feature in the present application is the vintage-year dependence of the score contributions. Assumptions 3 and 5 deliver the weak dependence and moment conditions needed to apply the random-field CLT (where compactness of Θ is inherited from Assumption 4). □

Corollary 2 (Structural asymptotic normality in the inception-only case)**.** *Under the additional conditions of Corollary 1,*

$$\sqrt{n}\left(\hat{\theta}_0 - \theta_0\right) \xrightarrow{d} \mathcal{N}(0, \Sigma_0).$$

Remark 4 (Interpretation of the asymptotic center)**.** *For $H > 0$, Proposition 4 is centered at $\theta_H^\dagger$, not at θ_0. Therefore, large-sample inference in the horizon-averaged case pertains to the pseudo-true parameter defined by the criterion itself. Structural interpretation relative to θ_0 requires the target-shift analysis developed later in the paper.*

3.4 Large sample inference

Building on the asymptotic normality established above, we now turn to the practical implementation of large-sample inference. Proposition 4 implies the familiar sandwich form

$$\Sigma_H = \mathcal{H}_H^{-1}\Lambda_H\mathcal{H}_H^{-1}.$$

We now describe consistent estimators of the two building blocks $\mathcal{H}_H$ and Λ_H.

Score contributions. It is useful to define the score contribution of unit i explicitly as

$$s_i^{(H)}(\theta) := \nabla_\theta L\left(\bar{\epsilon}_i^{(H)}(\theta)\right) = L'\left(\bar{\epsilon}_i^{(H)}(\theta)\right) \cdot \nabla_\theta \bar{\epsilon}_i^{(H)}(\theta). \tag{18}$$

For the quadratic loss $L(x) = x^2$, Equation (18) simplifies to

$$s_i^{(H)}(\theta) = 2\bar{\epsilon}_i^{(H)}(\theta)\nabla_\theta \bar{\epsilon}_i^{(H)}(\theta).$$

This explicit definition makes clear that the long-run covariance matrix Λ_H is the covariance of the criterion score, not the covariance of the pricing errors themselves.

Hessian estimator. A sample analogue of the Hessian is

$$\hat{\mathcal{H}}_H := \nabla_{\theta\theta} Q_{n,H}(\hat{\theta}_H) = -\frac{1}{n}\sum_{i=1}^{n} \nabla_{\theta\theta} L\Big(\bar{\epsilon}_i^{(H)}(\hat{\theta}_H)\Big). \tag{19}$$

When analytic derivatives are unavailable, the derivatives entering Equation (19) can be computed numerically. In practice, central finite differences are typically adequate, although automatic differentiation or quasi-Newton approximations may also be used. For a methodological paper, the exact numerical routine is secondary; what matters asymptotically is that $\hat{\mathcal{H}}_H$ consistently estimates $\mathcal{H}_H$. In our empirical applications, we use the central difference approximations of (Eu, 2017, Algorithm 2).

SHAC estimator of the long-run covariance matrix. Because the score contributions exhibit vintage-year dependence, a heteroskedasticity- and autocorrelation-consistent covariance estimator is required. We use the spatial HAC estimator of (Kim and Sun, 2011, Equation 2) adapted to the vintage-year metric:

$$\hat{\Lambda}_H = \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{n} k_{i,j}\, s_i^{(H)}(\hat{\theta}_H) s_j^{(H)}(\hat{\theta}_H)^\top, \qquad k_{i,j} = K\left(\frac{d_{i,j}}{b_V}\right), \tag{20}$$

where $d_{i,j}$ is the vintage-year distance from Assumption 3 and $K(\cdot)$ is a symmetric kernel satisfying the standard SHAC regularity conditions.

A convenient choice is the Bartlett kernel

$$K_{BT}(x) = \max(0, 1 - |x|).$$

The associated bandwidth sequence b_V should satisfy

$$b_V \to \infty, \qquad \frac{b_V}{V} \to 0, \tag{21}$$

which is the standard condition ensuring consistent long-run covariance estimation under weak dependence. Equation (21) is a population-theoretic requirement. In empirical work, one may implement a finite truncation rule for b_V and assess robustness across reasonable bandwidth choices.

This distinction between theory and implementation is important. A fixed bandwidth may be useful in a finite sample and can be justified as an approximation when dependence becomes negligible beyond a small vintage-year lag. However, under generic weak dependence, consistency of SHAC is established under a diverging bandwidth sequence such as Equation (21).

Asymptotic covariance matrix, standard errors, and Wald tests. The estimated asymptotic covariance matrix is

$$\widehat{\mathrm{Avar}}(\hat{\theta}_H) = \frac{1}{n}\hat{\mathcal{H}}_H^{-1}\hat{\Lambda}_H\hat{\mathcal{H}}_H^{-1}. \tag{22}$$

Accordingly, the vector of asymptotic standard errors is

$$\mathrm{SE}(\hat{\theta}_H) = \sqrt{\mathrm{diag}\left[\widehat{\mathrm{Avar}}(\hat{\theta}_H)\right]}. \tag{23}$$

This presentation is algebraically equivalent to the usual sandwich formula, but it is more transparent because the $1/n$ scaling is stated directly at the covariance-matrix level.

For linear hypotheses of the form

$$\text{Hypothesis}_0 : \mathcal{R}\theta = r, \qquad \text{Hypothesis}_1 : \mathcal{R}\theta \neq r,$$

the Wald statistic is

$$W_H = (\mathcal{R}\hat{\theta}_H - r)^\top \left[\mathcal{R}\widehat{\text{Avar}}(\hat{\theta}_H)\mathcal{R}^\top\right]^{-1} (\mathcal{R}\hat{\theta}_H - r), \tag{24}$$

which is asymptotically χ^2_q under Hypothesis_0, where q is the number of linear restrictions.

Remark 5 (Interpretation of inference when $H > 0$)**.** *When $H > 0$, the standard errors and Wald tests in Equations* (23) *and* (24) *are asymptotically valid for the pseudo-true parameter $\theta^\dagger_H$. They should therefore be interpreted as inference on the horizon-specific criterion minimizer, not automatically as structural tests about θ_0.*

3.5 Comparison to similar estimators

Our Least-Mean-Distance (LMD) estimator belongs to the broad class of semiparametric nonlinear extremum estimators in the sense of Pötscher and Prucha (1997). Its main advantage in the present setting is that it accommodates pooled fund-level pricing errors, flexible horizon averaging through $\mathcal{T}_{i,H}$, and increasing-domain asymptotics indexed by vintage years, all within a comparatively transparent objective function.

The relationship to the existing private-equity asset-pricing literature is best understood at the level of objective functions and aggregation schemes. The estimator is most closely connected to Driessen et al. (2012), while it is more loosely related to the time-series GMM construction of Korteweg and Nagel (2016). The relevant point is not that all three estimators are identical, but rather that they occupy nearby positions in the same semiparametric estimation landscape.

3.5.1 Connection to Driessen et al. (2012)

The connection to Driessen et al. (2012) is especially close. Under the inception-only restriction, equal normalization, and quadratic loss, the LMD criterion coincides with the identity-weighted one-step cross-sectional GMM objective used in their framework. To bridge the gap, let us rewrite the NPV_i from Equation 1 in our more general SDF notation as $PV_i(\theta) = \sum_{t\in\{t_{0,i},t_{0,i}+1,\dots\}} \Psi_{t_{0,i},t}(\theta)CF_t$.

Proposition 5 (Connection to the Driessen et al. (2012) objective)**.** *Suppose that:*

1. $\mathcal{T}_i = \{t_i^0\}$ *for all* i*,*
2. $w_i = 1$ *for all* i*, and*
3. $L(x) = x^2$*.*

Then the LMD estimator solves

$$\hat{\theta} = \arg\min_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^{n} (PV_i(\theta))^2 ,$$

which is equivalent to the identity-weighted cross-sectional GMM objective in Driessen et al. (2012).

Proof. Under $\mathcal{T}_i = \{t_i^0\}$, the averaged pricing error reduces to the inception-based pooled net present value

$$\bar{\epsilon}_i(\theta) = \epsilon_{t_i^0,i}(\theta) = \sum_{t=0}^{T} \frac{\Psi_t(\theta)}{\Psi_{t_i^0}(\theta)} CF_{i,t} = PV_i(\theta).$$

With $w_i = 1$ and $L(x) = x^2$, Equation (14) becomes

$$\hat{\theta} = \arg\min_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^{n} (PV_i(\theta))^2 .$$

Writing the vector of pooled pricing errors as $e(\theta) = (PV_1(\theta), \ldots, PV_n(\theta))^\top$, the objective is proportional to $e(\theta)^\top e(\theta)$, which is the identity-weighted one-step GMM criterion. □

3.5.2 Connection to Korteweg and Nagel (2016)

The relationship to Korteweg and Nagel (2016) is more indirect. Their estimator aggregates moments across funds first and then applies a quadratic form to the resulting time-series object. By contrast, the LMD criterion applies the loss function at the unit level and then averages across units. These two orderings of aggregation and loss evaluation are generally not equivalent.

For that reason, it is more accurate to describe Korteweg and Nagel (2016) as a closely related *time-series GMM analogue* rather than as a literal special case of Equation (14). The conceptual connection is still informative: both approaches estimate SDF parameters from pooled cash flows, but they differ in the population object they emphasize and in the source of asymptotic information.

Remark 6 (Why the distinction matters)**.** *The order in which pricing errors are aggregated is not innocuous. Applying the loss after aggregation emphasizes the behavior of a pooled moment vector, whereas applying the loss before aggregation emphasizes the cross-sectional distribution of unit-level pricing errors. As a result, the two criteria can differ in finite-sample robustness, identification strength, and asymptotic target.*

Empirically, Korteweg and Nagel (2016) chose to estimate the SDF parameters based on synthetic public market cash flows to use their estimated SDF model as Generalized Public Market Equivalent (GPME) benchmarking device for private fund-level cash flows.

3.5.3 Positioning of the present estimator

The contribution of the present estimator is therefore not that it dominates prior approaches in every dimension. Rather, its advantage is methodological fit for the current application: it combines a transparent nonlinear least-squares objective with explicit control over the discounting horizon, accommodates pooled non-traded cash flows at the unit level, and admits an increasing-domain asymptotic framework that maps naturally to vintage-year variation. Table 3 summarizes the most prominent distinctions between the three approaches.

	Driessen et al. (2012)	Korteweg and Nagel (2016)	Our approach
Estimator type	Cross-sectional NLS (one-step GMM)	Time-series GMM (public-market calibrated)	Nonlinear Least-Mean-Distance
Cash flows priced	PE fund cash flows	Public replicating strategies	PE fund cash flows
Discounting dates	Fund inception only	Fund inception only	Flexible via $\mathcal{T}_i$
Asymptotics	Infill ($n_i \to \infty$, N fixed)	$V \to \infty$	Increasing domain ($V \to \infty$, n_v bounded)
Inference	Bootstrap	Spatial HAC	Spatial HAC
Cross-sectional unit	Vintage year portfolio	Single fund	Both (tested empirically)
SDF specification	Linear factor model	Exponentially affine	Both (tested empirically)

Table 3: Comparison of SDF estimation frameworks for private equity.

4 Implementation choices and simulation

In this section and the next, we move from the general methodology to our empirical application. First, we describe the data and establish our sample restriction based on terminal cash-flow measurement concerns. Second, we detail the empirical implementation choices that anchor the general estimator from Section 3.2. Third, we use a simulation study to validate and calibrate these choices. Finally, Section 5 presents the resulting empirical estimates for Buyout and Venture Capital funds.

4.1 Private data

We use the Preqin cash flow dataset as of 14th April 2022, which is well known in the academic private equity literature (Harris et al., 2014; Brown et al., 2015; Korteweg and Nagel, 2016; Ang et al., 2018). To keep the empirical analysis clear and concise, we filter the dataset for Buyout (BO) and Venture Capital (VC) funds. We only keep funds with fund currency US Dollar (USD) and region equal to North

America. In rare instances, the Preqin taxonomy assigns geographic labels that are inconsistent with the North America region filter (e.g., a fund classified under a sub-regional or country-level tag that nonetheless passes the broader filter). Because all retained funds are denominated in USD, these classification inconsistencies do not introduce foreign-exchange contamination.

For this subset of BO and VC funds, we extract all (i) equal-weighted (hereafter EW) and (ii) value-weighted (hereafter FW, for fund-size-weighted) cash flow series (cf. Figures 2 and 3). In the overall dataset, the minimum vintage year is 1982, and the maximum is 2021. We initially retain all vintage years, including recent ones. Section 4.3 assesses the impact of unseasoned fund Net Asset Values (NAVs) on our empirical estimation results. On this basis, we decide how many recent vintage years to remove to reduce reliance on unrealized NAVs. For the vintage years we keep, all liquidated and non-liquidated funds enter our estimation. For non-liquidated funds, we treat the latest NAV as final cash flow.

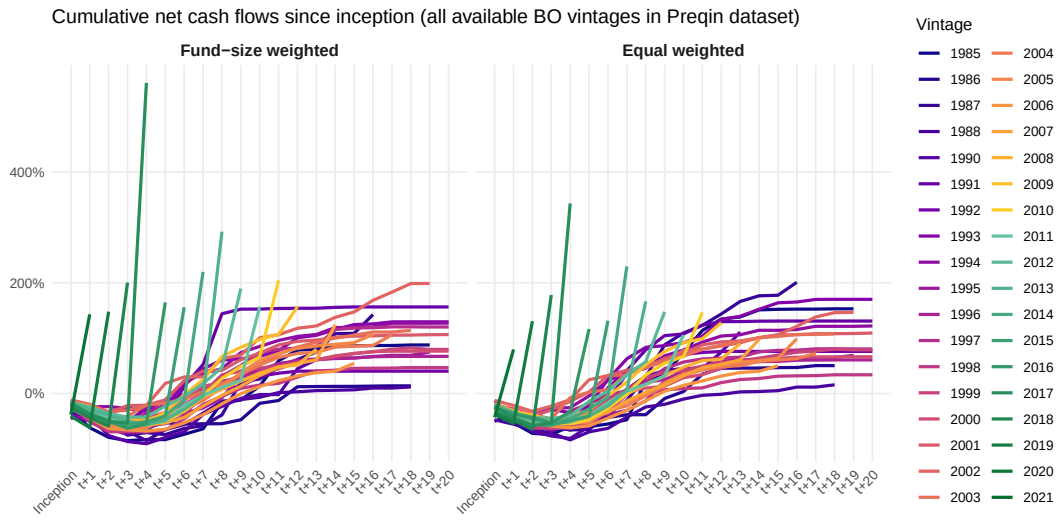

Figure 2: **Cumulative cash flows:** J-curves by vintage year for BO funds in Preqin dataset scaled by fund commitment. Post-2010 vintage years are shown as green lines. Based on the sensitivity analysis in Section 4.3, these vintages are excluded from the empirical estimates in Section 5 to limit reliance on unrealized NAVs.

Buyout sample. The Buyout (North America) sample contains 933 distinct funds spanning 36 vintage years. The vintage year distribution is as follows: 1985 (2), 1986 (3), 1987 (5), 1988 (6), 1990 (2), 1991 (3), 1992 (6), 1993 (5), 1994 (14), 1995 (11), 1996 (13), 1997 (14), 1998 (32), 1999 (27), 2000 (34), 2001 (21), 2002 (14), 2003 (21), 2004 (24), 2005 (44), 2006 (50), 2007 (49), 2008 (45), 2009 (19), 2010 (31), 2011 (25), 2012 (39), 2013 (36), 2014 (44), 2015 (35), 2016 (57), 2017 (35), 2018 (45), 2019 (65), 2020 (34), 2021 (23). The region distribution is as follows: Americas (4), Asia (16), Canada (1), Central and East Europe (2), Europe (6),

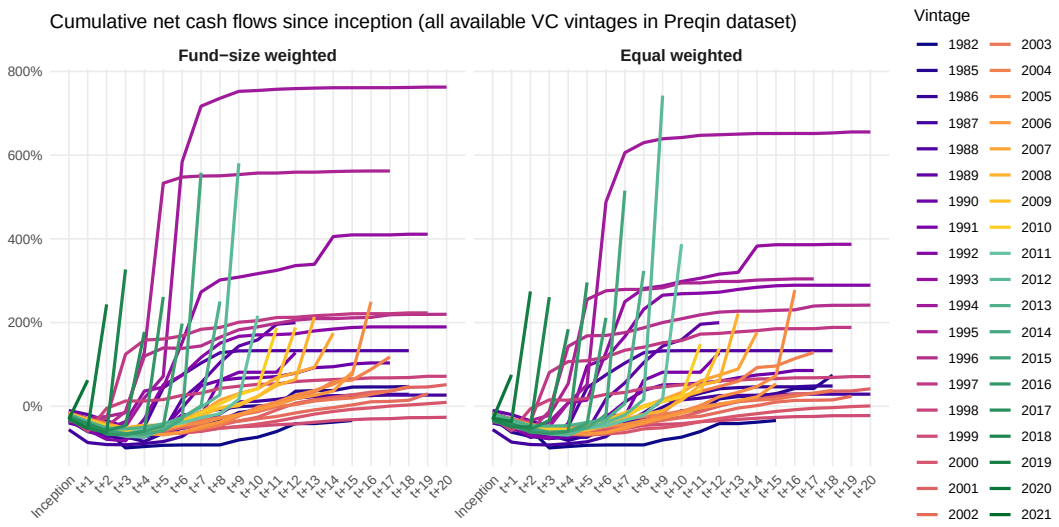

Figure 3: **Cumulative cash flows:** J-curves by vintage year for VC funds in Preqin dataset scaled by fund commitment. Post-2010 vintage years are shown as green lines. Based on the sensitivity analysis in Section 4.3, these vintages are excluded from the empirical estimates in Section 5 to limit reliance on unrealized NAVs.

Japan (1), Midwest (2), North America (461), South America (2), Southeast (1), Southwest (1), US (433), West Europe (1). The strategy distribution is as follows: Buyout (933). The fund currency distribution is as follows: USD (933).

Venture Capital sample. The Venture Capital (North America) sample contains 968 distinct funds spanning 38 vintage years. The vintage year distribution is as follows: 1982 (1), 1985 (3), 1986 (4), 1987 (2), 1988 (1), 1989 (1), 1990 (4), 1991 (1), 1992 (7), 1993 (6), 1994 (6), 1995 (8), 1996 (14), 1997 (16), 1998 (28), 1999 (44), 2000 (70), 2001 (45), 2002 (28), 2003 (18), 2004 (31), 2005 (34), 2006 (44), 2007 (48), 2008 (38), 2009 (18), 2010 (18), 2011 (23), 2012 (21), 2013 (32), 2014 (33), 2015 (37), 2016 (40), 2017 (50), 2018 (52), 2019 (64), 2020 (45), 2021 (33). The region distribution is as follows: Asia (10), Australasia (1), Canada (1), China (10), Europe (3), India (2), Mexico (1), Midwest (2), North America (267), Northeast (2), Southwest (1), Thailand (1), US (664), West (1). The strategy distribution is as follows: Early Stage (274), Early Stage: Seed (47), Early Stage: Start-up (26), Expansion / Late Stage (127), Venture (General) (472), Venture Debt (22). The fund currency distribution is as follows: USD (968).

4.2 SDF models and public data

We test a simple linear SDF model similar to Driessen et al. (2012)

$$\Psi^{\mathrm{SL}}_{\tau,t}(\theta) = \frac{\prod_{h=0}^{\tau} \ (1+\alpha+r_{f,h}+\sum_m \ \beta_m \ F_{m,h})}{\prod_{h=0}^{t} \ (1+\alpha+r_{f,h}+\sum_m \ \beta_m \ F_{m,h})} \tag{25}$$

and an exponential affine SDF model adapted from Korteweg and Nagel (2016)

$$\Psi_{\tau,t}^{\mathrm{EA}}(\theta) = \exp\left[-\sum_{h=\tau+1}^{t}\left(\alpha + \log(1+r_{f,h}) + \sum_{m\in M}\beta_m \cdot \log(1+F_{m,h})\right)\right] \tag{26}$$

with monthly (arithmetic) risk-free return r_f, monthly (arithmetic) zero-net-investment portfolio returns F_m, and parameter vector $\theta = (\alpha, \beta)$. Throughout both the simulation and the empirical application, all factor returns and fund cash flows are observed at monthly frequency.

The public market factors that enter our SDF draw on the US dataset of the recently popularized q^5 investment factor model sourced from `http://global-q.org/factors.html` (Hou et al., 2015, 2021). Their five-factor model includes the market excess return (MKT), a size factor (ME), an investment factor (IA), a return on equity factor (ROE), and an expected growth factor (EG). The maximum date in the q^5 investment factor dataset is 2024-12-31.

To avoid overfitting, we restrict our analysis to six simple SDF models that contain {MKT} alone or {MKT} plus {ME or IA or ROE or EG or Alpha}. In Equation 14, we use the quadratic loss function $L(x) = x^2$.

Remark 7 (Alpha Bounds)**.** *In our optimization procedure, we limit the Alpha factor α to the range of $[-2\%, +2\%]$ per month to avoid the "exploding alpha" issue documented in (Driessen et al., 2012, Footnote 8): "Note that one needs to put an upper bound on α when performing the optimization in equation (3). This is to exclude the possibility that α tends to infinity." In this degenerate scenario, an infinite alpha can almost perfectly price any cash flow stream (except its first component), rendering the beta risk factors irrelevant.*

4.3 The impact of unseasoned funds and reported NAVs

Figure 4 presents the vintage-year cutoff sensitivity analysis separately for BO and VC under fund-value weighting (FW). A clear upward-drift pattern is present in both sub-samples: as more recent, unseasoned vintages are included, the estimated market beta increases monotonically. For BO under value weighting, $\hat{\beta}_{\mathrm{MKT}}$ at the 0-year Horizon (see Remark 8) rises from approximately 1.23 with a 2011 vintage cutoff to roughly 2.42 with the full sample (2021). Notably, most of the increase is concentrated in vintages after 2016, where $\hat{\beta}_{\mathrm{MKT}}$ jumps sharply from 1.53 (2016 cutoff) to 2.34 (2017 cutoff), reflecting the disproportionate influence of young, unrealized funds. For VC, the drift is similarly dramatic: the FW estimate at the 0-year Horizon climbs from approximately 1.57 (2011 cutoff) to 2.33 with the full sample. Discounting all final NAVs by 50% substantially flattens the upward trend: the BO FW estimate range compresses to 1.09–1.65, and the VC FW range to 1.41–1.54, confirming that the drift is attributable to the outsized mechanical reliance on subjectively reported Net Asset Values for young funds. These findings reinforce the sample restriction to the 2010 maximum vintage year applied throughout the remainder of this section.

Remark 8 (Definition of Horizon)**.** *Throughout the empirical section, we use the term* Horizon *H to denote the maximum compounding horizon in years used in*

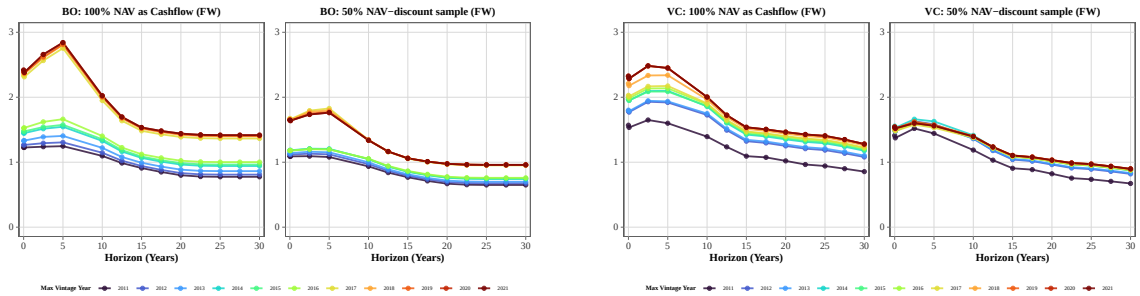

Figure 4: **Sensitivity analysis (MKT factor):** Single-factor model for BO (left) and VC (right) under value weighting, using VYP with different vintage-year cutoffs (asymptotic estimates). For each asset class, the left chart shows the standard baseline (full NAV as final cash flow), while the right chart applies a 50% discount to all final NAVs.

the set of discounting dates $\mathcal{T}$ (cf. Equation 13). Formally, starting from the fund inception date t_i^0, estimating a model with a Horizon H implies that the estimator minimizes the average pricing error over the set of discounting dates $\mathcal{T}_{i,H} = \{t_i^0, t_i^0 + 1, \ldots, \min(t_i^0 + 12H, T_i^{\max})\}$, formally denoted as $\bar{\epsilon}_i^{(H)}$. Note that the Horizon has as upper bound the max cash flow date per fund or portfolio in our empirical application. This Horizon in years serves as x-axis in all subsequent empirical and simulation charts.

4.4 NAV discount sensitivity analysis

To further quantify the sensitivity of factor loading estimates to terminal Net Asset Value treatment, we conduct a systematic NAV discount analysis on the seasoned sample (maximum vintage year 2010). Figure 5 reports single-factor MKT estimates under five NAV treatments: the baseline (full reported NAV as final cash flow), 25%, 50%, and 75% constant discounts applied to all final NAVs, and a complete NAV write-off (100% discount).

For BO, the estimated market beta decreases monotonically with the NAV discount. At the 0-year Horizon under EW, $\hat{\beta}_{\text{MKT}}$ drops from 1.26 (no discount) to 1.19 (25% discount), 1.12 (50%), 1.03 (75%), and 0.94 (100% write-off); under FW, the corresponding range is 1.20 to 0.98. At the 15-year Horizon, the EW estimate ranges from 0.81 (no discount) to 0.55 (100% write-off), while the FW estimate spans 0.84 to 0.62. The qualitative conclusion is robust across all NAV treatments: the single-factor market model remains economically plausible and stable.

For VC under EW, the pattern is analogous but the sensitivity is more muted in percentage terms due to the higher baseline beta: the 0-year estimate declines from 1.88 (no discount) to 1.84 (25%), 1.80 (50%), 1.77 (75%), and 1.74 (100%); at the 15-year Horizon, from 1.30 (no discount) to 1.09 (100% write-off). Under FW, the sensitivity is more pronounced, particularly at longer Horizons: the 0-year estimate ranges from 1.59 to 1.34, while the 15-year estimate declines from 1.07 to 0.73.

This analysis demonstrates that terminal NAV treatment has a material but

bounded effect on estimated factor loadings. The monotonic relationship confirms the intuition that reported NAVs mechanically inflate estimated betas: funds whose final cash flow is a subjective valuation rather than a realized exit price appear more systematically risky because their NAVs partially co-move with public market indices. Importantly, even under the extreme assumption of a complete NAV write-off, the first qualitative conclusions of this subsection can be summarized as follows: BO maintains a market beta near or below unity, while VC retains a substantially higher systematic exposure. We do not impose a model-based residual-value adjustment, as this would constitute a separate research agenda (Jenkinson et al., 2020; Brown et al., 2023). Instead, the range spanned by the five NAV treatments provides a transparent envelope of estimation uncertainty attributable to terminal value measurement.

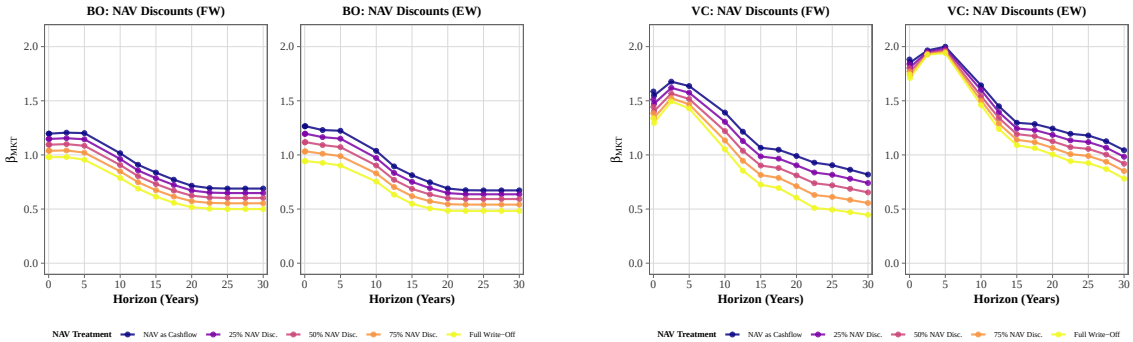

Figure 5: **NAV discount sensitivity (MKT factor):** Single-factor model for BO (left) and VC (right) using VYP and maximum vintage year 2010. Each panel shows asymptotic estimates under five NAV treatments: full NAV as final cash flow, 25%, 50%, 75% constant discounts, and 100% NAV write-off.

4.5 Simulation study

Our Monte Carlo experiments examine the bias and variance of our estimation methodology in finite samples.[3] Specifically, we investigate five key dimensions of the estimation framework: (i) the finite-sample efficiency gains of forming vintage-year portfolios versus using individual funds; (ii) the tradeoff between expanding the time-series dimension (number of vintage years) and the cross-sectional dimension (funds per vintage); (iii) the robustness of the estimates to the functional specification of the SDF (simple linear vs. exponential affine); (iv) the behavior of multivariate identification under idiosyncratic noise using two-factor models; and (v) as our main overarching topic, the optimal choice of the maximum compounding Horizon (i.e., the set of discounting dates $\mathcal{T}$).

[3]As with any simulation study, it investigates the ability to identify the assumed data generating process rather than validating the corresponding SDF model itself.

We use historical q-investment factors from 1986 to 2005 and simulate 20 funds for each of these 20 vintage years. This means we condition our simulation analysis on realized q-investment factors paths (see the Macro-Conditioning Kernel discussion in Section 3.1). Each fund contains 15 deals with equal investment amounts and exactly one divestment cash flow. Deals are entered within the first five years of the fund lifetime following a discrete uniform distribution and afterward held between one to ten years again uniformly distributed. The deal returns are generated by the simple linear or exponential affine SDF models described in Equations 25 and 26. In the base case, we just use the MKT factor with $\beta_{\text{MKT}} = 1$ and in each month, add a normal i.i.d. error term with monthly standard deviation $\sigma = 0.2$ (annualized variance $12 \cdot 0.2^2 = 0.48$, corresponding to an annualized volatility of $0.2\sqrt{12} \approx 69\%$) and zero mean. In the exponential affine case, we adjust the lognormally distributed error mean to zero by subtracting $0.5\sigma^2$. If a negative return exceeds -100%, the company defaults with a zero exit cash flow. In contrast, the error term in the simulations of Driessen et al. (2012) is more well-behaved as it follows a shifted lognormal distribution that, even with arbitrarily high error term variance, just allows for returns below say -99%, if the market return is close to its lower bound (see equation 9 in their online appendix). We specifically analyze the remaining divergence from their simulation design in Appendix B.1. In our simulation scenarios, we evaluate the estimator across a full grid of compounding Horizons, where the maximum month in the set of discounting dates $\mathcal{T}$ ranges from 0 up to 30 years (0-360 months). To assess our estimator's bias and variance, we simulate 1000 test scenarios for both vintage-year and single-fund portfolios.

Cross-sectional unit i: As anticipated in Subsection 3.3, vintage year portfolios demonstrate lower bias and variance than individual funds. Under the simple linear SDF and a 15-year Horizon (maximum month 180), the mean (standard deviation) of the coefficient estimate $\hat{\beta}_{\text{MKT}}$ is 0.966 (0.188) for vintage year portfolios, compared to 0.887 (0.359) for individual funds. The superiority of the portfolio approach is especially striking when evaluating the entire Horizon curve in Figure 6. Estimates for individual funds are highly unstable, exhibiting severe upward bias at short Horizons ($\hat{\beta}_{\text{MKT}} \approx 1.5$) and deteriorating to negative values at long Horizons. In contrast, vintage year portfolios yield substantially more stable estimates that remain within a narrow, economically plausible range around the true value of 1.0. The portfolio approach consistently achieves a lower standard deviation across all simulated Horizons, confirming that grouping funds effectively mitigates idiosyncratic noise.

This trade-off between bias/variance reduction and sample size has important implications. While forming vintage year portfolios substantially improves the accuracy of the estimates, it dramatically reduces the number of cross-sectional units, thereby weakening the relevance of asymptotic approximations. This constraint likely explains the choice of Korteweg and Nagel (2016) to rely on individual funds as cross-sectional units in their asymptotic SHAC framework, enabling them to exploit a larger sample size for smaller standard errors.

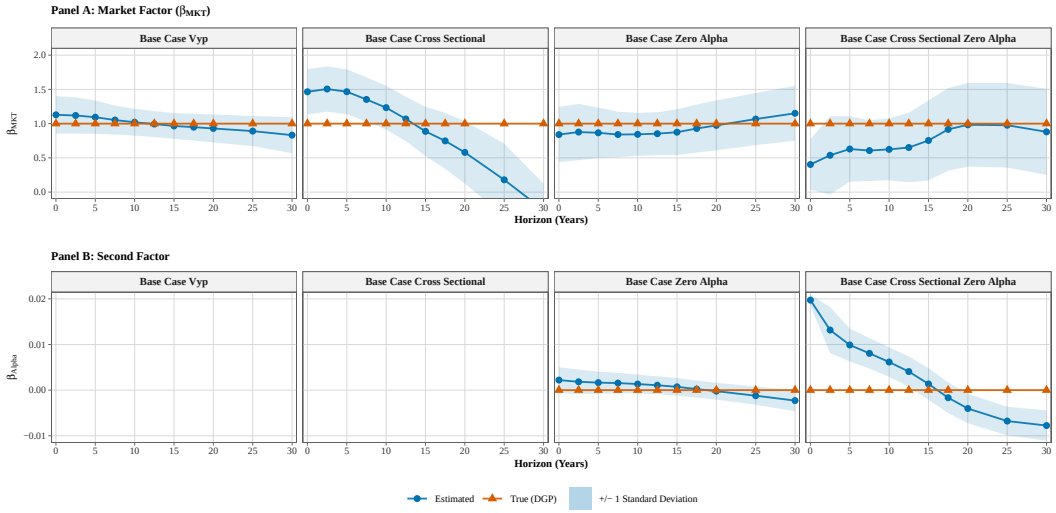

Figure 6: **Simulation ($\beta_{\mathbf{MKT}}$, α):** Single-fund vs. vintage-year portfolios.

Varying vintages V and portfolio sizes n/V: A fundamental question in private equity panel data econometrics is whether precision is best improved by extending the time-series dimension (V, number of vintage years) or the cross-sectional dimension (n/V, number of funds per vintage portfolio). To assess this tradeoff, we compare five data configurations (depicted in Figures 7 and 8): (1) our base case for this experiment ($V = 20$, $n/V = 40$); (2) an extended vintage sample with small portfolios ($V = 39$, $n/V = 10$); (3) an extended vintage sample with medium portfolios ($V = 39$, $n/V = 20$); (4) an early-vintage sub-sample ($V = 10$, years 1986–1995, $n/V = 20$); and (5) a late-vintage sub-sample ($V = 10$, years 1996–2005, $n/V = 20$).

In our simulation design, increasing the number of cross-sectional units per portfolio (n/V) is substantially more effective at suppressing estimator variance than adding more vintage years (V). For the single-factor model at the 15-year Horizon, transitioning from 10 funds to 20 funds per vintage over a 39-year period (scenarios 2 vs. 3) reduces the standard deviation of $\hat{\beta}_{\text{MKT}}$ by roughly 22%, from 0.218 to 0.169. In contrast, shrinking the time-series dimension from 39 years to just 10 years (while holding $n/V = 20$ constant) elevates the variance only moderately for the early-vintage sub-sample ($\sigma(\hat{\beta}_{\text{MKT}}) \approx 0.194$ for 1986–1995), though the late-vintage sub-sample (1996–2005) exhibits considerably higher variance ($\sigma(\hat{\beta}_{\text{MKT}}) \approx 0.432$), reflecting the sensitivity to the realized macro environment. This hierarchy of variance reduction is even more pronounced in the two-factor setting, where idiosyncratic cash flow noise severely impairs multivariate identification: increasing n/V aggressively dampens the spurious α variance. Consequently, our findings provide an empirical rationale for the finite-sample superiority of the Driessen et al. (2012) infill asymptotics (fixing V, letting $n_v \to \infty$) over approaches that rely purely on time-series aggregation.

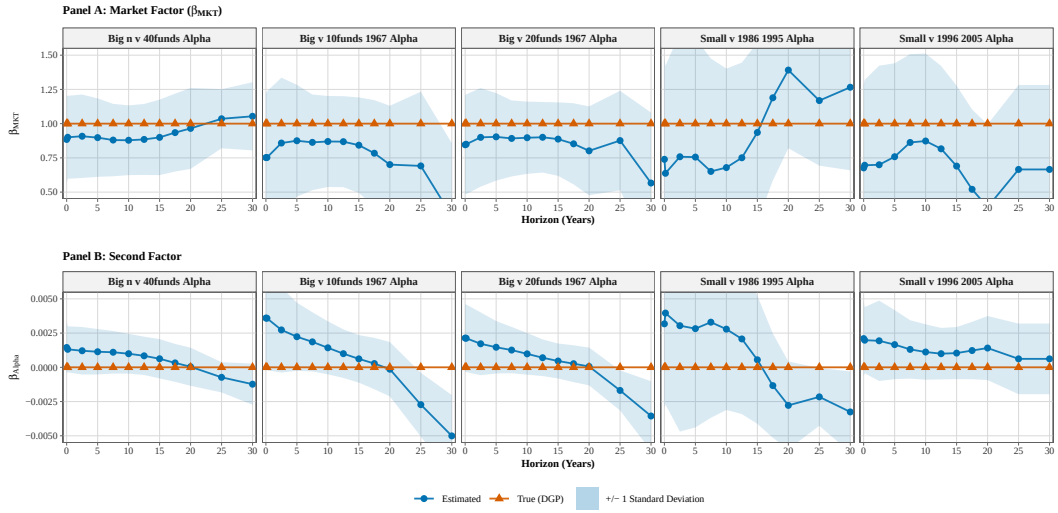

Figure 7: **Simulation ($\beta_{\mathbf{MKT}}$, α):** Varying vintages V and portfolio sizes n/V.

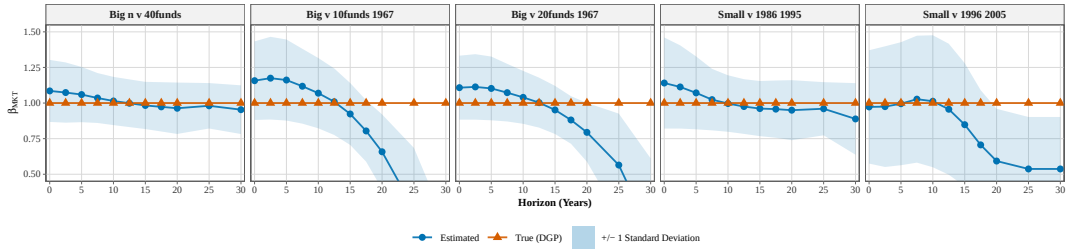

Figure 8: **Simulation ($\beta_{\mathbf{MKT}}$):** Varying vintages V and portfolio sizes n/V.

SDF model Ψ: In Figure 9, we compare the simple linear SDF model against the exponential affine specification. For the single-factor model, the bias is minimized at the 12.5-year Horizon (150 months), where both functional forms perform almost identically and yield estimates extremely close to the true value of 1.0: the exponential affine yields a mean market beta of 0.999 (0.182), matching the simple linear's 0.994 (0.189). For the two-factor (MKT+Alpha) model at this same 12.5-year Horizon, both functional forms again yield nearly identical parameter results, though with an attenuated market beta ($\hat{\beta}_{\text{MKT}} \approx 0.85$) and a minor spurious positive alpha ($\hat{\alpha} \approx 0.001$). In the two-factor case, the market beta bias is actually minimized later, around the 20-year Horizon ($\hat{\beta}_{\text{MKT}} \approx 0.97$, $\hat{\alpha} \approx -0.0002$).

However, a salient difference emerges at extremely short compounding Horizons. At a 1-month Horizon (resembling an NPV-only discounting framework), the exponential affine formulation deteriorates much more severely in both models, manifesting a larger upward bias (e.g., in the single-factor case, $\hat{\beta}_{\text{MKT}} = 1.21$ vs. 1.13) and elevated variance (0.39 vs. 0.27) compared to the linear model. Overall, the perceived superiority of exponential affine SDFs (Gourieroux and Monfort, 2007;

Bertholon et al., 2008) does not translate into empirical finite-sample advantages in our framework. The simple linear model actually appears slightly better behaved.

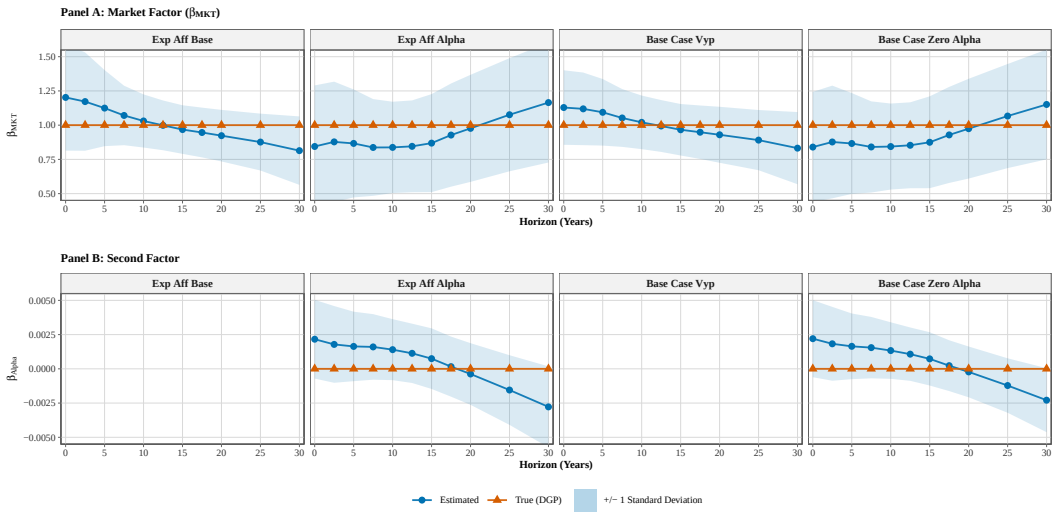

Figure 9: **Simulation ($\beta_{\mathbf{MKT}}$, α):** Exponential affine vs. linear models.

Two-factor models (q^5 factors): Figure 10 reports simulation results for two-factor models that pair MKT with each of the four non-market q^5 factors (ME, IA, ROE, EG), with true loadings $\beta_{\text{MKT}} = 1$ and $\beta_{\text{2nd}} = 0.5$. A common pattern emerges across all four specifications: the second-factor loading is substantially overestimated at short Horizons and attenuated toward (or beyond) the true value at long Horizons, while the market factor absorbs a compensating bias in the opposite direction. Estimation variance is markedly larger than in the single-factor case, reflecting the collinearity between correlated factors in a limited cross-section of vintage year portfolios. Table 4 summarizes the key estimates at the shortest Horizon and at 15 years, which is approximately the MSE-minimizing Horizon for most models; we highlight the most notable factor-specific features below.

The ME model is the only specification in which $\hat{\beta}_{\text{2nd}}$ undergoes a sign change (at roughly 17.5 years), declining from 0.86 to -0.02 over the full Horizon range, a severe attenuation that would qualitatively alter economic conclusions. The IA model exhibits a large second-factor bias at short Horizons ($\hat{\beta}_{\text{IA}} \approx 1.06$ vs. truth 0.5) but both loadings converge close to their true values near the 17.5-year Horizon ($\hat{\beta}_{\text{MKT}} = 0.97$, $\hat{\beta}_{\text{IA}} = 0.52$). The EG model exhibits the largest overall estimation variance ($\sigma(\hat{\beta}_{\text{EG}}) \approx 1.5$ at the shortest Horizon) but achieves a reasonably accurate joint estimate around 15–17.5 years ($\hat{\beta}_{\text{MKT}} = 0.91$–$1.02$, $\hat{\beta}_{\text{EG}} = 0.47$–$0.56$). The ROE model is the exception: both loadings remain persistently biased at all Horizons, with $\hat{\beta}_{\text{MKT}}$ never exceeding 0.91 and $\hat{\beta}_{\text{ROE}}$ never falling below 0.62. These results suggest that empirical researchers should exercise caution when interpreting two-factor SDF estimates for private equity, and that the MSE-minimizing Horizon is model-dependent, typically falling in the 12.5–17.5-year range.

Model	$\hat{\beta}_{\text{MKT}}$		$\hat{\beta}_{\text{2nd}}$		$\sigma(\hat{\beta}_{\text{MKT}})$		$\sigma(\hat{\beta}_{\text{2nd}})$	
	0 yr	15 yr	0 yr	15 yr	0 yr	15 yr	0 yr	15 yr
ME + MKT	1.15	0.97	0.87	0.59	0.28	0.21	0.46	0.40
IA + MKT	0.97	0.92	1.06	0.64	0.34	0.30	0.74	0.53
ROE + MKT	0.88	0.83	0.98	0.69	0.43	0.49	0.61	0.49
EG + MKT	0.87	0.91	0.99	0.56	0.61	0.45	1.53	0.31

True values: $\beta_{\text{MKT}} = 1$, $\beta_{\text{2nd}} = 0.5$.

Table 4: Two-factor simulation summary: mean estimates and standard deviations at the shortest (0 yr) and the 15-year Horizon.

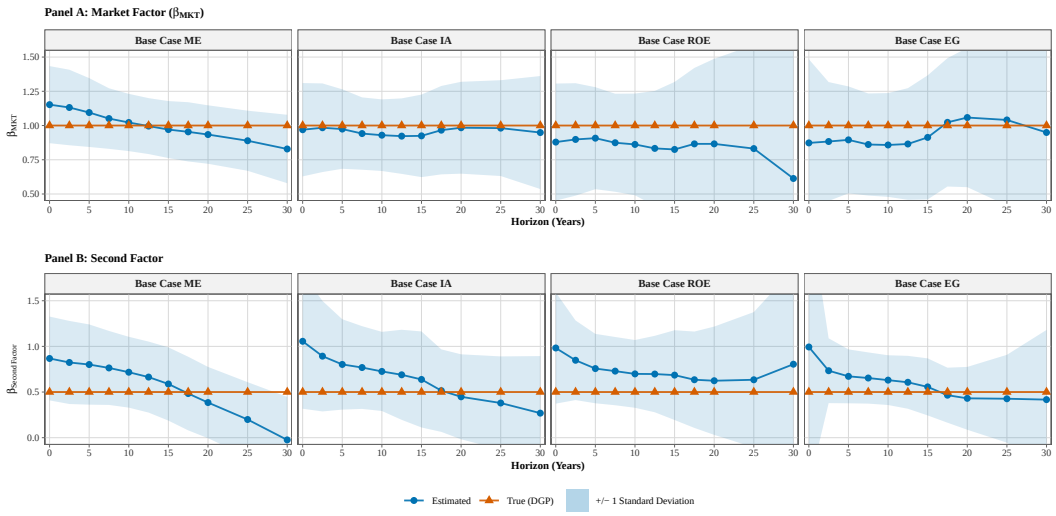

Figure 10: **Simulation (q^5 factors):** Two-factor models using VYP.

Varying fund lifetimes: Figure 11 illustrates how the optimal Horizon shifts in response to abbreviated fund lifetimes. As anticipated, the intercept of the alpha parameter decreases for shorter fund lifetimes relative to our baseline scenario of a five-year investment period and a ten-year maximum holding period. Specifically, for the shortest examined fund lifetime of two years (comprising a one-year investment period and a one-year maximum holding period), the beta bias is minimized at a 12.5-year Horizon, while the alpha bias is minimized at a 5-year Horizon. Similarly, for a fund with a two-year investment period and a two-year maximum holding period, the minimum beta bias occurs at a 15-year Horizon, and the minimum alpha bias at a 7.5-year Horizon. Consequently, these results demonstrate that for funds with short lifespans, the optimal Horizon extends significantly beyond the fund's actual lifetime.

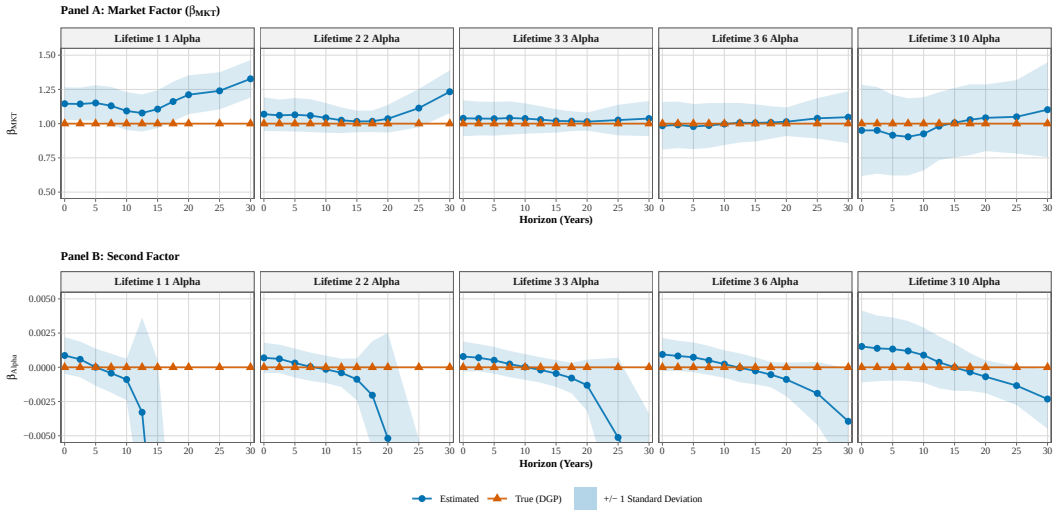

Figure 11: **Simulation ($\beta_{\mathbf{MKT}}$, α):** Varying fund lifetimes using VYP with the following yearly investment-period/holding-period pairs: 1/1, 2/2, 3/3, 3/6, and 3/10 (from left to right).

Size of set $\mathcal{T}$ (Maximum Compounding Horizon): Our most critical simulation finding relates to the optimal choice of the maximum compounding Horizon, i.e., the size of the set of discounting dates $\mathcal{T}$. From a purely theoretical perspective (cf. Section 3.1), discounting all cash flows exclusively to the fund inception date, corresponding to a zero-month or NPV-only Horizon in our notation, should yield unconditionally unbiased estimates under the deal-level exact-pricing restriction (Equation (2)). This minimal choice for $\mathcal{T}$ corresponds exactly to the established Driessen et al. (2012) framework.

However, our diverse simulation results reveal a puzzling disconnect between asymptotic theory and finite-sample performance: extending the set of discounting dates deep into the fund lifecycle drastically improves the estimator's precision. Across almost all model specifications (see Figures 10 and 9), bias is minimized not at the theoretical 0-year Horizon, but rather around a 12.5 to 15-year compounding Horizon. At extremely short Horizons, idiosyncratic cash flow noise interacts nonlinearly with the estimation procedure, inducing severe bias in factor loadings. By averaging pricing errors over multiple future dates, effectively incorporating the Timing Risk Premium, our approach acts as a powerful finite-sample bias correction.

Interestingly, this "variance reduction technique" improves the mean estimate far more than it reduces standard errors. While estimator variance traces a shallow U-shape with a minimum near the 15-year Horizon (see Table 4) for fund with a lifetime of 15 years, this reduction is secondary to the striking stabilization of the parameter bias. However, our simulations for shorter fund lifetimes indicate that determining the optimal horizon requires tailored simulation designs rather than a simple heuristic assuming the optimal horizon always coincides with the fund's life. We further analyze these puzzling results from a theoretical perspective in Appendix A, detailing the *bias–variance tradeoff* and the role of the Timing Risk Premium.

Simulation conclusion: To conclude, our simulation study rationalizes two key practices from the Driessen et al. (2012) paper: (i) vintage-year portfolio formation helps to improve estimator precision, and (ii) increasing the number of funds per vintage seems more effective in controlling estimator variance than increasing the number of vintages (for i.i.d. error terms). Our second finding provides an empirical rationale for the choice of Driessen et al. (2012) to employ infill asymptotics that let the number of underlying funds per vintage tend to infinity. However, our correctly specified scenarios cannot support the assumption of Korteweg and Nagel (2016) that (iii) the exponential affine SDF is superior to the simple linear SDF in a multi-period framework; in fact, their finite-sample bias and variance profiles are remarkably similar. Our simulation study also suggests that (iv) averaging pricing errors over multiple dates substantially reduces the severe short-Horizon bias inherent to the original procedure of Driessen et al. (2012), which discounts all cash flows exclusively to the fund inception date. Ultimately, selecting a set of compounding dates $\mathcal{T}$ that spans a significant portion of the fund lifecycle appears to decrease finite-sample bias (and to a lesser extent, variance) more effectively than all other measures combined. Appendix A, in particular Section A.6, explains the theoretical foundation of this observation by showing how NFV-induced timing terms can shift the population target while still improving finite-sample performance through the bias–variance tradeoff.

Accordingly, in the empirical application we use vintage-year portfolios, the simple linear SDF, emphasize one-factor MKT specifications, treat two-factor results as exploratory, and focus on Horizons around 12.5–15 years while also reporting sensitivity over a broader grid.

4.6 Role of asymptotic inference

To assess the parameter significance in the empirical application, we compute the asymptotic standard errors as outlined in Subsection 3.4. For the Bartlett kernel's bandwidth $b_n = D$ we select 12 years, i.e., funds with absolute vintage year differences larger than 12 years are assumed to be independent. The asymptotic results provide the benchmark large-sample approximation.

At the same time, private-equity samples remain small in the vintage-year and cross-sectional dimensions, and model choice is often data-driven. For that reason, the empirical analysis complements asymptotic inference with out-of-sample validation and stability diagnostics. To account for the dependency between funds from adjacent vintage years caused by overlapping fund cash flows, we draw on hv-block cross-validation (Racine, 2000). As Racine (2000) demonstrates, hv-block cross-validation is a consistent model-selection criterion for dependent data: the h-block gap between training and validation sets ensures approximate independence, while the v-block provides a sufficiently large validation sample within each fold. In our implementation, each cross-validation fold partitions the data into three disjoint groups: (i) a *validation set* (v-block) comprising funds from three consecutive vintage years (e.g., 2000, 2001, 2002), on which the out-of-sample pricing error is evaluated; (ii) a *buffer zone* (h-block) of three vintage years on each side of the v-block (e.g., 1997–1999 and 2003–2005), which is discarded entirely to break the

dependence between training and validation data caused by overlapping fund cash flows; and (iii) an *estimation set* consisting of all remaining vintage years (e.g., 1985–1996 and 2006–end), on which the model parameters are estimated. This three-way split is repeated across rolling vintage-year windows, as detailed in Table 5. The number of active folds adapts dynamically to the selected maximum vintage year cutoff in any given estimation run. For instance, when estimating the full sample through 2021, we apply ten folds; when estimating a heavily seasoned sample truncated at 2010 to avoid reliance on subjective interim NAVs, the framework naturally truncates to seven active folds.

Remark 9 (Cross-validation as stability diagnostic)**.** *The hv-block cross-validation quantities reported throughout this paper serve as* ***stability diagnostics and model-selection criteria*** *rather than inferential standard errors. They summarize the fold-to-fold variation in parameter estimates under dependence-reducing sample splits and are informative about model sensitivity and effective parameter curvature, but they do not provide a valid null distribution for the full-sample estimator. For formal statistical inference (standard errors and hypothesis tests), we rely on the SHAC-based asymptotic framework of Section 3.4. In practice, hv-block cross-validation complements the asymptotic analysis by (i) diagnosing model stability and out-of-sample performance without relying on asymptotics and (ii) flagging over-parameterized specifications that appear well-identified in-sample but collapse out-of-sample.*

training.before	h-block.before	v-block	h-block.after	training.after
estimation	remove	validation	remove	estimation
start-1984	1985,1986,1987	1988,1989,1990	1991,1992,1993	1994-end
start-1987	1988,1989,1990	1991,1992,1993	1994,1995,1996	1997-end
start-1990	1991,1992,1993	1994,1995,1996	1997,1998,1999	2000-end
start-1993	1994,1995,1996	1997,1998,1999	2000,2001,2002	2003-end
start-1996	1997,1998,1999	2000,2001,2002	2003,2004,2005	2006-end
start-1999	2000,2001,2002	2003,2004,2005	2006,2007,2008	2009-end
start-2002	2003,2004,2005	2006,2007,2008	2009,2010,2011	2012-end
start-2005	2006,2007,2008	2009,2010,2011	2012,2013,2014	2015-end
start-2008	2009,2010,2011	2012,2013,2014	2015,2016,2017	2018-end
start-2011	2012,2013,2014	2015,2016,2017	2018,2019,2020	2021-end

Table 5: Partitions used for hv-block cross-validation. The number of active folds truncates dynamically based on the maximum vintage year cutoff of the specific estimation run.

5 Empirical results for BO and VC data

This section presents the empirical SDF estimates for Buyout (BO) and Venture Capital (VC) funds. As the simulation study in Subsection 4.5 confirms, we use

vintage-year portfolios (VYPs) to estimate simple linear SDF models for the BO and VC sub-samples individually. We restrict the data to North American, USD-denominated funds from the Preqin dataset described in Subsection 4.1. This restriction ensures consistency between fund cash flows and the U.S. q-investment factors used throughout the analysis. Moreover, we only consider funds with a maximum vintage year of 2010, to reduce the reliance on unrealized NAVs (see Section 4.3.

We analyze BO and VC as two distinct asset classes with fundamentally different risk profiles and report both equal-weighted (EW) and value-weighted (FW) estimates across a grid of compounding Horizons. Our baseline specification is the single-factor MKT model estimated on vintage-year portfolios. Two-factor models that add an alpha intercept or a single q-investment factor serve as robustness and exploratory extensions.

For each model specification, we report asymptotic standard errors (SE_A) from the SHAC framework alongside hv-block cross-validation fold dispersions (SD_{CV}), defined as the standard deviation of parameter estimates across hold-out folds. Throughout the empirical analysis, t_A denotes the t-ratio based on asymptotic standard errors; SR_{CV} (stability ratio) denotes the ratio of the full-sample estimate to the cross-validation fold dispersion, i.e., $\mathrm{SR}_{CV} := \hat{\theta}/\mathrm{SD}_{CV}$. As discussed in Remark 9, SR_{CV} complements asymptotic inference as a stability diagnostic.

All R code and data (including the previous simulations) are available in an online repository: `https://github.com/quant-unit/Fundwise_SDF/tree/master/r_project`.

5.1 Baseline: single market-factor models

Tables 6–7 and Figure 12 report the single-factor MKT model for BO and VC across Horizons from 0 to 30 years. As anticipated by the simulation study, the market beta declines with the compounding Horizon for both asset classes.

For BO under EW, $\hat{\beta}_{\mathrm{MKT}}$ ranges from approximately 1.22–1.27 at short Horizons (0–5 years) and decreases to 0.67 at 25–30 years; under FW, the corresponding range is 1.20–1.21 down to 0.69. The divergence between asymptotic and cross-validation diagnostics persists: asymptotic t-ratios (t_A) are moderate at short Horizons (e.g., 1.10 under EW at 0 years) and decline further at longer Horizons, while the cross-validation stability ratios remain high throughout (SR_{CV} ranging from 3.75 to 10.65 under EW and from 2.63 to 8.68 under FW), indicating robust stability across data subsets.

For VC, the market beta is substantially higher across all Horizons, consistent with venture capital's greater sensitivity to systematic risk. Under EW, $\hat{\beta}_{\mathrm{MKT}}$ reaches approximately 1.97–2.00 at 2.5–5 years and converges toward 1.04–1.13 beyond 25 years; under FW, the corresponding range is 1.64–1.68 down to 0.82–0.86. Despite the higher parameter variability reflected in larger asymptotic standard errors (SE_A of 1.2–1.8), the cross-validation stability diagnostics indicate robust estimates (SR_{CV} of 3.59–4.46 under EW; 3.54–7.01 under FW), although somewhat lower than the BO stability ratios due to the more heterogeneous VC fund universe. The converged long-horizon estimates (BO $\approx$ 0.67, VC $\approx$ 1.0–1.1 under

EW) are economically plausible: BO funds, which invest in mature companies with stable cash flows, exhibit market exposure near or slightly below unity, while VC funds, which back early-stage ventures with highly uncertain outcomes, retain a substantially higher beta even at seasoned Horizons.

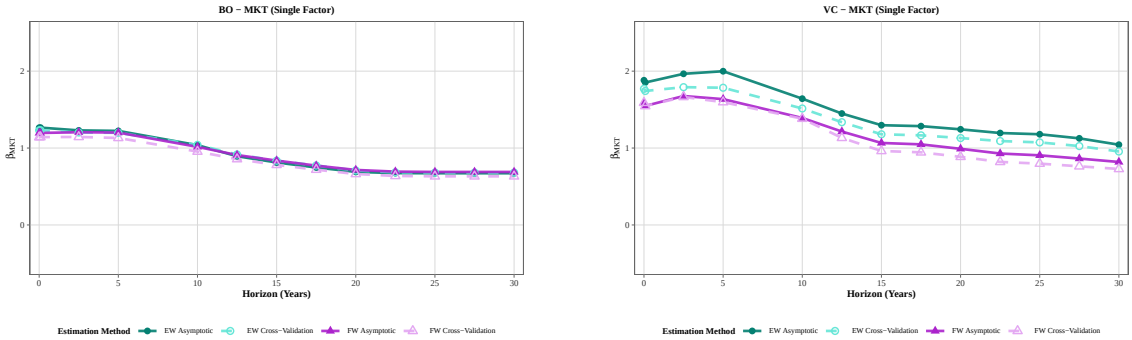

Figure 12: **Empirical (MKT factor):** Single-factor model for BO (left) and VC (right) using VYP and maximum vintage year 2010.

Table 6: Empirical Estimates: BO — Single-Factor (MKT) Model

	Equal-Weighted (EW)					Value-Weighted (FW)				
Horizon	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}
0 yr	1.264	1.154	0.123	1.10	10.00	1.197	1.170	0.139	1.02	8.23
0.1 yr	1.265	1.155	0.116	1.10	10.65	1.196	1.170	0.132	1.02	8.68
2.5 yr	1.229	0.975	0.133	1.26	9.07	1.206	1.175	0.170	1.03	6.73
5 yr	1.223	0.972	0.147	1.26	8.16	1.201	1.173	0.196	1.02	5.78
10 yr	1.038	1.642	0.189	0.63	5.52	1.015	1.806	0.228	0.56	4.19
12.5 yr	0.894	1.660	0.224	0.54	4.08	0.909	1.627	0.247	0.56	3.46
15 yr	0.812	1.511	0.223	0.54	3.75	0.837	1.798	0.246	0.47	3.18
17.5 yr	0.747	1.405	0.206	0.53	3.75	0.771	1.699	0.248	0.45	2.90
20 yr	0.690	2.001	0.163	0.34	4.36	0.715	1.737	0.245	0.41	2.71
22.5 yr	0.674	1.947	0.136	0.35	5.03	0.693	1.687	0.241	0.41	2.65
25 yr	0.672	1.942	0.131	0.35	5.18	0.689	1.678	0.240	0.41	2.63
27.5 yr	0.672	1.942	0.131	0.35	5.18	0.689	1.678	0.240	0.41	2.63
30 yr	0.672	1.942	0.131	0.35	5.18	0.689	1.678	0.240	0.41	2.63

Notes: $\hat{\beta}$ = parameter estimate; SE_A = asymptotic standard error; SD_{CV} = cross-validation fold dispersion (standard deviation across hold-out folds); t_A = asymptotic t-ratio; SR_{CV} = stability ratio (estimate / fold dispersion). Significance markers based on t_A: $^{*}\, p < 0.05$, $^{**}\, p < 0.01$, $^{***}\, p < 0.005$.

5.2 Robustness: two-factor $\alpha + MKT$ models

Tables 8–9 and Figure 13 present the two-factor MKT + α model for BO and VC. The results show the pathological behavior anticipated by the simulation study: once an unconstrained alpha intercept is introduced, the estimation destabilizes and the MKT loading collapses.

Table 7: Empirical Estimates: VC — Single-Factor (MKT) Model

	Equal-Weighted (EW)					Value-Weighted (FW)				
Horizon	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}
0 yr	1.881	1.765	0.402	1.07	4.40	1.586	1.645	0.359	0.96	4.42
0.1 yr	1.852	1.756	0.390	1.06	4.46	1.548	1.651	0.335	0.94	4.62
2.5 yr	1.965	1.803	0.451	1.09	3.97	1.677	1.646	0.453	1.02	3.67
5 yr	1.999	1.824	0.493	1.10	3.62	1.636	1.643	0.437	1.00	3.65
10 yr	1.642	1.273	0.364	1.29	4.15	1.391	1.783	0.318	0.78	4.35
12.5 yr	1.449	1.269	0.304	1.14	4.39	1.214	2.005	0.151	0.61	7.49
15 yr	1.297	1.215	0.264	1.07	4.46	1.066	6.793	0.137	0.16	7.01
17.5 yr	1.285	1.216	0.279	1.06	4.17	1.048	6.123	0.148	0.17	6.36
20 yr	1.243	1.220	0.283	1.02	4.00	0.990	4.655	0.159	0.21	5.55
22.5 yr	1.195	1.320	0.272	0.91	4.02	0.929	4.906	0.176	0.19	4.66
25 yr	1.179	1.314	0.272	0.90	3.95	0.905	4.420	0.183	0.20	4.36
27.5 yr	1.126	1.292	0.269	0.87	3.81	0.863	3.749	0.194	0.23	3.93
30 yr	1.043	1.600	0.266	0.65	3.59	0.819	3.220	0.206	0.25	3.54

Notes: $\hat{\beta}$ = parameter estimate; SE_A = asymptotic standard error; SD_{CV} = cross-validation fold dispersion (standard deviation across hold-out folds); t_A = asymptotic t-ratio; SR_{CV} = stability ratio (estimate / fold dispersion). Significance markers based on t_A: $^{*}\,p < 0.05$, $^{**}\,p < 0.01$, $^{***}\,p < 0.005$.

For BO, the alpha term absorbs a substantial share of the explanatory power: under EW, $\hat{\alpha}$ ranges from 0.003 to 0.006 per month and the MKT coefficient oscillates erratically between 0.41 and 1.01, with no stable convergence across Horizons; under FW, $\hat{\alpha}$ is persistently around 0.005–0.008 per month while the MKT loading collapses to values between 0.10 and 0.32. The asymptotic standard errors are highly erratic, with several entries at or very near zero interspersed with values exceeding 25. Cross-validation fold dispersions for $\hat{\beta}_{\mathrm{MKT}}$ are large (0.09–0.49 under EW; 0.65–0.74 under FW), confirming that the collapsed point estimates are unstable across data subsets.

For VC, the two-factor model exhibits a distinctive and economically puzzling pattern. At short Horizons (0–0.1 years under EW), the alpha term reaches its upper bound ($\hat{\alpha} = 0.020$/month) and the MKT coefficient drops to approximately 0.47, resembling the BO and PE behavior. However, as the Horizon increases beyond 5 years, the MKT loading rises steadily to values exceeding 2.3, while the alpha estimate turns progressively negative, reaching -0.009/month beyond 25 years. This reversal is unique to VC and likely reflects the extreme idiosyncratic risk inherent to early-stage ventures: at long Horizons, the optimizer increasingly attributes the high compounded returns of surviving VC funds to market co-movement rather than alpha, while the negative alpha compensates for selection effects among the funds that underperformed. Under FW, the instability is somewhat attenuated ($\hat{\alpha}$ remains close to zero at short Horizons; $\hat{\beta}_{\mathrm{MKT}} \approx 1.73$), but the overall fragility persists, with asymptotic standard errors exceeding 7–11 across most Horizons.

The BO and VC two-factor results therefore corroborate the conclusion drawn for PE: single-factor MKT models are the most statistically robust specification for private equity SDF estimation in our sample, given the limited cross-section of vintage-year portfolios.

Table 8: Empirical Estimates: BO — MKT + Alpha Model

	Equal-Weighted (EW)					Value-Weighted (FW)				
Horizon	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}
Panel A: Market Factor (β_{MKT})										
0 yr	1.005	1.066	0.489	0.94	1.19	0.147	25.778	0.711	0.01	0.16
0.1 yr	1.005	1.069	0.430	0.94	1.48	0.168	24.142	0.716	0.01	0.19
2.5 yr	0.446	1.379	0.377	0.32	1.55	0.230	6.788	0.739	0.03	0.26
5 yr	1.003	2.078	0.087	0.48	11.22	0.322	6.558	0.707	0.05	0.43
10 yr	0.522	40.420	0.298	0.01	2.18	0.376	3.337	0.652	0.11	0.67
12.5 yr	1.000	3.644	0.386	0.27	1.59	0.277	5.044	0.651	0.05	0.60
15 yr	0.406	0.385	0.357	1.06	1.53	0.231	0.453	0.682	0.51	0.52
17.5 yr	0.995	2.953	0.364	0.34	2.08	0.230	11.801	0.707	0.02	0.50
20 yr	0.488	0.788	0.435	0.62	1.50	0.140	14.154	0.723	0.01	0.43
22.5 yr	0.495	0.779	0.451	0.64	1.47	0.097**	0.035	0.726	2.75	0.39
25 yr	0.508	0.769	0.447	0.66	1.51	0.096**	0.031	0.726	3.08	0.39
27.5 yr	0.508	0.769	0.447	0.66	1.51	0.096**	0.031	0.726	3.08	0.39
30 yr	0.508	0.769	0.447	0.66	1.51	0.096**	0.031	0.726	3.08	0.39
Panel B: Alpha (β_{Alpha})										
0 yr	0.004	0.006	0.004	0.74	1.54	0.008	0.192	0.005	0.04	1.50
0.1 yr	0.004	0.006	0.003	0.73	1.58	0.008	0.179	0.005	0.05	1.46
2.5 yr	0.006	0.014	0.003	0.45	1.83	0.007	0.055	0.005	0.13	1.30
5 yr	0.003	0.013	0.002	0.22	1.45	0.006	0.052	0.005	0.12	1.18
10 yr	0.004	0.363	0.002	0.01	1.27	0.005	0.019	0.004	0.26	0.87
12.5 yr	-0.000	0.124	0.003	-0.00	0.95	0.005	0.036	0.005	0.14	0.81
15 yr	0.003	0.002	0.002	1.88	0.99	0.005	0.003	0.005	1.59	0.73
17.5 yr	-0.001	0.161	0.003	-0.01	0.21	0.004	0.103	0.005	0.04	0.62
20 yr	0.002	0.023	0.003	0.07	0.30	0.005	0.134	0.005	0.03	0.63
22.5 yr	0.002	0.023	0.003	0.07	0.26	0.005***	0.000	0.005	10.88	0.66
25 yr	0.001	0.024	0.003	0.06	0.23	0.005***	0.000	0.005	10.84	0.66
27.5 yr	0.001	0.024	0.003	0.06	0.23	0.005***	0.000	0.005	10.84	0.66
30 yr	0.001	0.024	0.003	0.06	0.23	0.005***	0.000	0.005	10.84	0.66

Notes: $\hat{\beta}$ = parameter estimate; SE_A = asymptotic standard error; SD_{CV} = cross-validation fold dispersion (standard deviation across hold-out folds); t_A = asymptotic t-ratio; SR_{CV} = stability ratio (estimate / fold dispersion). Significance markers based on t_A: $^{*}\, p < 0.05$, $^{**}\, p < 0.01$, $^{***}\, p < 0.005$.

Table 9: Empirical Estimates: VC — MKT + Alpha Model

	Equal-Weighted (EW)					Value-Weighted (FW)				
Horizon	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}
Panel A: Market Factor (β_{MKT})										
0 yr	0.473	18.860	0.187	0.03	3.91	1.732	11.028	0.585	0.16	2.40
0.1 yr	0.458	21.158	0.201	0.02	3.86	1.723***	0.237	0.575	7.26	2.40
2.5 yr	1.427	1.806	0.373	0.79	3.67	1.556	9.260	0.341	0.17	4.62
5 yr	1.475	2.296	0.326	0.64	4.44	1.454	8.251	0.323	0.18	4.77
10 yr	1.796	25.246	0.190	0.07	9.49	1.879	17.174	0.237	0.11	8.44
12.5 yr	1.970	4.329	0.205	0.46	9.57	2.002	10.569	0.159	0.19	13.06
15 yr	2.129	6.375	0.441	0.33	4.46	2.119	7.601	0.155	0.28	13.94
17.5 yr	2.224	5.140	0.243	0.43	9.01	2.164	7.186	0.195	0.30	11.27
20 yr	2.284	4.663	0.265	0.49	8.43	2.182	6.896	0.231	0.32	9.53
22.5 yr	2.304	4.669	0.266	0.49	8.41	2.184***	0.316	0.248	6.91	8.81
25 yr	2.303	4.669	0.270	0.49	8.29	2.179***	0.323	0.243	6.75	9.02
27.5 yr	2.315	4.607	0.271	0.50	8.31	2.189***	0.283	0.237	7.73	9.28
30 yr	2.326	4.542	0.268	0.51	8.46	2.193***	0.272	0.234	8.05	9.41
Panel B: Alpha (β_{Alpha})										
0 yr	0.020	0.115	0.009	0.17	1.61	-0.001	0.048	0.010	-0.03	0.45
0.1 yr	0.020	0.130	0.009	0.15	1.46	-0.001	0.001	0.010	-1.84	0.43
2.5 yr	0.007	0.009	0.005	0.81	1.35	0.001	0.043	0.003	0.03	0.60
5 yr	0.006	0.012	0.004	0.54	1.14	0.002	0.039	0.003	0.05	0.37
10 yr	-0.001	0.119	0.002	-0.01	-1.30	-0.004	0.095	0.001	-0.04	-4.27
12.5 yr	-0.004	0.006	0.001	-0.78	-5.86	-0.006	0.059	0.001	-0.11	-7.52
15 yr	-0.007	0.036	0.002	-0.18	-2.99	-0.008	0.048	0.001	-0.16	-7.27
17.5 yr	-0.007	0.032	0.001	-0.23	-9.95	-0.008	0.046	0.001	-0.18	-6.11
20 yr	-0.008	0.030	0.001	-0.27	-8.85	-0.008	0.045	0.002	-0.19	-5.30
22.5 yr	-0.008	0.030	0.001	-0.28	-8.18	-0.009***	0.001	0.002	-9.67	-4.97
25 yr	-0.008	0.030	0.001	-0.28	-8.14	-0.009***	0.001	0.002	-9.61	-5.07
27.5 yr	-0.008	0.029	0.001	-0.29	-8.45	-0.009***	0.001	0.002	-9.74	-5.22
30 yr	-0.009	0.029	0.001	-0.29	-8.66	-0.009***	0.001	0.002	-9.81	-5.36

Notes: $\hat{\beta}$ = parameter estimate; SE_A = asymptotic standard error; SD_{CV} = cross-validation fold dispersion (standard deviation across hold-out folds); t_A = asymptotic t-ratio; SR_{CV} = stability ratio (estimate / fold dispersion). Significance markers based on t_A: $^{*}\,p < 0.05$, $^{**}\,p < 0.01$, $^{***}\,p < 0.005$.

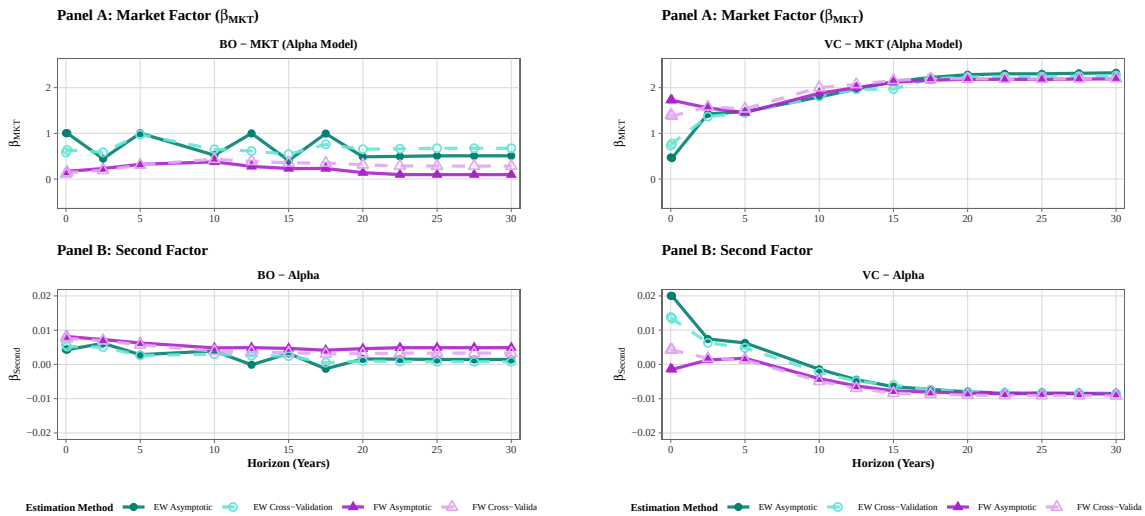

Figure 13: **Empirical ($MKT + \alpha$ model):** Two-factor model for BO (left) and VC (right) using VYP and maximum vintage year 2010.

5.3 Two-factor q^5-factor models

We now pair MKT with each of the four non-market q^5 factors (ME, IA, ROE, EG) in two-factor specifications estimated separately for BO and VC. Because the detailed estimation tables contain extensive Horizon-by-Horizon results, we do not reproduce them here and instead summarize the key patterns visible in Figures 14 and 15; the full tables are available in the accompanying online repository.

Buyout. Figure 14 shows a uniform pattern across the four non-market factors. At short Horizons (0–5 years), the second-factor loadings are positive and economically large: the ME loading reaches approximately 1.4–1.6 under EW (with stability ratios SR_{CV} of 3.0–3.8), the IA loading starts near 1.0 under EW ($\mathrm{SR}_{CV} \approx 1.4$), ROE near 0.6 ($\mathrm{SR}_{CV} \approx 0.9$), and EG near 0.5 ($\mathrm{SR}_{CV} \approx 1.3$). As the Horizon increases, however, all four second-factor loadings decay toward zero and frequently change sign beyond 10–15 years. The ME factor retains a moderate positive loading even at long Horizons under FW ($\hat{\beta}_{\mathrm{ME}} \approx 1.3$, $\mathrm{SR}_{CV} \approx 2.1$), but under EW the loading drops below 1.0 after 17.5 years. The EG, IA, and ROE loadings all fluctuate around zero at intermediate and long Horizons, with cross-validation stability ratios well below conventional thresholds. The MKT loading in these two-factor models remains similar to its single-factor counterpart (ranging from roughly 1.0–1.1 at short Horizons to 0.63–0.76 at long Horizons), showing that a second factor does not materially alter the market risk assessment. The asymptotic standard errors are erratic throughout, often exceeding 10 and occasionally reaching triple-digit values, which highlights the unreliability of asymptotic inference in this setting.

Venture Capital. Figure 15 displays qualitatively analogous but more extreme patterns, consistent with the higher idiosyncratic risk inherent to early-stage ven-

tures. At short Horizons, the second-factor loadings are strikingly large: the IA loading reaches 5.1 under EW at the 0-year Horizon ($\mathrm{SR}_{CV} \approx 1.4$) and 4.2 under FW, the ROE loading starts at 1.5 (EW) and 1.0 (FW), the EG loading at approximately 3.5 (EW) and 6.6 (FW), while the ME factor is the exception with a subdued loading near zero or slightly positive (0.3 under EW, -0.03 under FW). All four second-factor loadings undergo dramatic sign reversals as the Horizon extends. The IA loading, for instance, turns negative beyond 10 years and eventually reaches -1.5 to -1.9 at long Horizons with large stability ratios (SR_{CV} exceeding -12). Similarly, EG and ROE produce large negative loadings at long Horizons ($\hat{\beta}_{\mathrm{EG}} \approx -0.9$, $\hat{\beta}_{\mathrm{ROE}} \approx -1.2$ under EW at 30 years). These sign reversals echo the behavior documented in the simulation study (Section 4.5), where two-factor models exhibited severe attenuation and sign changes in the second-factor loading across Horizons. The MKT loading in the VC two-factor models varies substantially across specifications: in the IA model it stabilizes around 1.4–1.5 at intermediate Horizons, while in the ROE and EG models it ranges from 0.4 to 2.0 depending on the Horizon and weighting scheme.

Summary. The q^5-factor results for BO and VC confirm the fragility of multi-factor SDF estimation documented in the preceding MKT + α analysis. While some second-factor loadings attain statistical significance at isolated Horizons (particularly ME for BO and short-Horizon IA for VC), none of the four non-market factors produces a stable, economically interpretable loading across the full Horizon spectrum. The pervasive sign reversals and the extreme sensitivity of point estimates to the compounding Horizon indicate that these results are driven by finite-sample noise rather than genuine systematic risk exposures. We conclude that the q^5-factor two-factor models do not improve upon the single-factor MKT benchmark for either asset class.

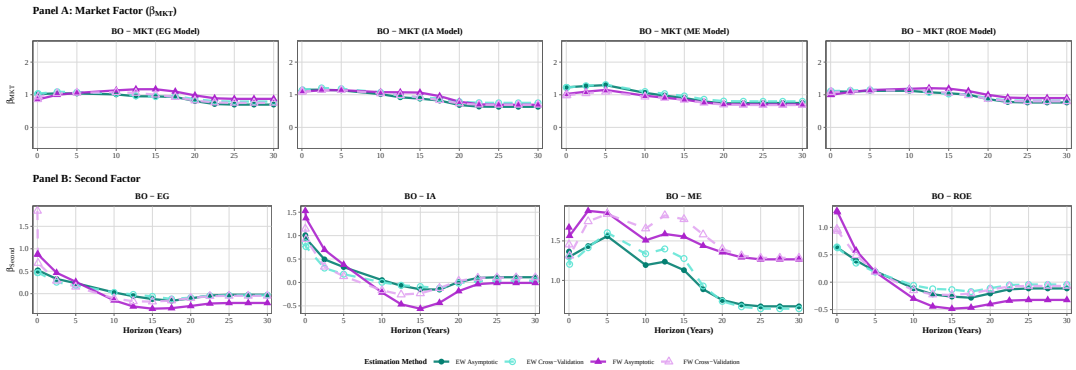

Figure 14: **Empirical results (q^5-factors):** Two-factor model for BO funds using VYP and maximum vintage year 2010.

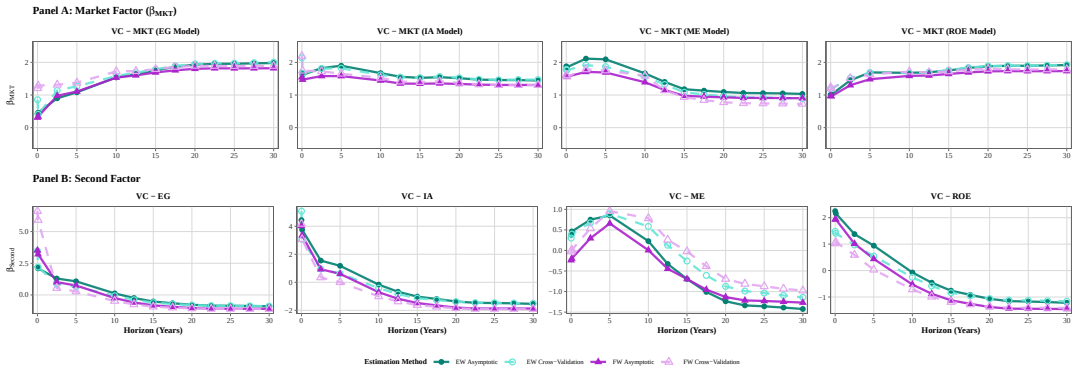

Figure 15: **Empirical results (q^5-factors):** Two-factor model for VC funds using VYP and maximum vintage year 2010.

5.4 Interpretation

The empirical results for Buyout and Venture Capital funds demonstrate consistent model reliability challenges across both asset classes and weighting schemes.

Single-factor MKT models are the most robust specification. The single-factor market model produces economically plausible and statistically significant estimates for both BO and VC. For Buyout, the market beta ranges from approximately 1.27 at the shortest Horizon to 0.67 at long Horizons under equal weighting, with cross-validation stability ratios consistently exceeding 3.7. For Venture Capital, the corresponding range is 1.88–2.00 down to 1.04–1.13, reflecting the substantially higher systematic risk exposure of early-stage investments. The cross-validation stability diagnostics are strong and consistent across the entire Horizon spectrum, even as asymptotic standard errors fluctuate erratically. The divergence between asymptotic inference and cross-validation diagnostics mirrors the simulation finding that, with the limited cross-section of vintage-year portfolios, the asymptotic approximation is unreliable and overly conservative.

Multi-factor models are fragile. Both the MKT + α and the q^5-factor two-factor models collapse as soon as a second parameter is introduced. In the MKT + α specification, the alpha intercept absorbs explanatory power from MKT, producing collapsed or negative market betas and erratic standard errors. The q^5-factor models exhibit a pervasive pattern of sign reversals: second-factor loadings are consistently overestimated at short Horizons and attenuate toward or beyond zero at long Horizons, exactly as predicted by the simulation study in Section 4.5. This fragility is not confined to a single factor or asset class; it appears across all eight specifications (four factors $\times$ two asset classes), ruling out the chance that an individual q^5 factor might robustly augment the market model.

BO–VC decomposition. Analyzing Buyout and Venture Capital separately provides a more informative test of model stability than pooling all private equity

funds into a single sample. BO and VC represent fundamentally different investment strategies: Buyout targets mature companies with leveraged balance sheets and stable cash flows, while Venture Capital finances early-stage firms with highly uncertain, option-like payoff profiles. Despite these differences, both sub-samples exhibit the same diagnostic pattern: robust single-factor significance, alpha-induced destabilization, and second-factor fragility. The fact that this pattern appears across two asset classes with different risk characteristics suggests that the multi-factor instability is a finite-sample identification problem, not an artifact of pooling heterogeneous fund types.

Reconciliation with the simulation study. The empirical results mirror the simulation predictions from Section 4.5. First, the MSE-minimizing Horizon at which the single-factor estimates stabilize (approximately 15–20 years) aligns with the 12.5–15-year optimum identified in simulation. Second, the sign reversals and large estimation variance observed in the empirical q^5-factor models replicate the pathological behavior documented in the two-factor simulation experiments, where the second-factor loading underwent severe attenuation and occasionally changed sign across Horizons. Third, the divergence between asymptotic inference and cross-validation stability diagnostics in the data is consistent with the simulation finding that asymptotic standard errors systematically overstate true parameter uncertainty in small samples. These results show that the finite-sample challenges identified by our simulation study appear directly in empirical estimation with real-world private market data.

Implications for empirical practice. Our findings yield a clear recommendation for applied researchers and practitioners. In our sample, semiparametric SDF estimation for private equity cash flows is substantially more reliable when restricted to single-factor market models estimated on seasoned, vintage-year portfolios at an extended compounding Horizon of approximately 15 years. Multi-factor models and alpha intercepts, while theoretically appealing, cannot be reliably estimated with the cross-sectional sample sizes currently available in private equity databases. Only a substantial expansion of the available data, either through longer time series of fully seasoned funds or, even better, through larger cross-sections within each vintage year, would provide the statistical power necessary to disentangle multiple systematic risk exposures from finite-sample noise.

6 Conclusion

Our generalized Least-Mean-Distance framework provides a flexible theoretical foundation for estimating Stochastic Discount Factor (SDF) models using non-traded cash flows. However, our Monte Carlo simulations reveal that semiparametric estimators in this setting face severe finite-sample challenges. Specifically, we demonstrate that the foundational Driessen et al. (2012) estimator exhibits substantially greater bias and variance than previously recognized, particularly concerning the estimation of abnormal performance (α). Our framework shows that this finite-sample

bias can be dramatically mitigated by averaging pricing errors across multiple future dates spanning the fund lifecycle, while estimator variance is most effectively controlled by increasing the number of cross-sectional units (n).

In the data-sparse private equity domain, where estimation relies on a limited cross-section of vintage-year portfolios, asymptotic approximations prove unreliable. We therefore strongly advocate for the use of robust resampling or cross-validation stability diagnostics tailored to overlapping cash flow dependencies. While relying on individual fund data yields smaller asymptotic standard errors, our simulations indicate that this approach exacerbates finite-sample bias and estimation noise, rendering it an unsuitable remedy. Given that virtually all multi-factor models in our empirical analysis fail to yield consistently stable parameter estimates under either asymptotic inference or cross-validation stability diagnostics, and this fragility is replicated independently for both Buyout and Venture Capital funds despite their markedly different risk profiles, we conclude that, in our sample, single-factor MKT specifications are substantially more stable than multi-factor alternatives and should be preferred (without α terms) until significantly broader cross-sections of private equity data become available. Researchers must exercise caution even then, ensuring that data-driven model selection (e.g., via hv-block cross-validation) does not yield economically implausible outcomes.

Expanding the SDF framework to robustly accommodate multiple risk factors requires imposing additional economic or statistical structure. Future research could further explore parametric assumptions for the cash flow generating process (Ang et al., 2018) or extract intra-period information from reported Net Asset Values (Brown et al., 2023). Methodologically, addressing the high estimator variance demands statistical techniques suited for sparse, high-dimensional settings. Approaches such as LASSO, ridge regression, or model averaging to regularize non-market coefficients, as well as ensemble learning methods that combine multiple weak learners, present promising paths forward.

Another vital avenue for future work involves explicitly modeling financial engineering, complex fee structures and carried interest within the data generating process. This would enable the estimation of gross-versus-net-of-fee SDFs, providing sharper evaluation models for both General Partners (GPs) and Limited Partners (LPs). Such structural refinements would naturally improve parameter accuracy and allow for a more rigorous analytical characterization of finite-sample bias. Additionally, carefully integrating intermediate NAVs as proxies for unobservable true valuations could help alleviate the timing risk premium bias inherent to terminal value concepts, though this requires downweighting actual cash distributions in favor of subjective accounting metrics.

Finally, we highlight the critical role of the compounding Horizon in empirical asset pricing for private markets. Reporting estimates on a future-value-based horizon grid yields an additional array of alternative parameter estimates that can be compared to the traditional NPV-based results of the Driessen et al. (2012) method. Our simulation study demonstrates that utilizing this future-value framework, i.e., averaging pricing errors over a broad set of discounting dates $\mathcal{T}$, yields vastly superior finite-sample properties compared to traditional net present value approaches. The simulation bias is minimized when the set of discounting dates corresponds

approximately to the typical fund lifecycle. Developing a formal theoretical or empirical mechanism to endogenously determine the optimal configuration of $\mathcal{T}$ for a given fund sample and macroeconomic environment remains an important objective for future research.

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A Bias–Variance Tradeoff of $\mathcal{T}$-Averaging

This appendix clarifies why $\mathcal{T}$-averaging can improve finite-sample performance even though, for $H > 0$, it generally shifts the population target away from the structural parameter θ_0. The key mechanism is a tradeoff between: (i) a *population target shift* induced by NFV terms through the Timing Risk Premium (TRP), and (ii) a *finite-sample stabilization effect* from averaging imperfectly correlated pricing errors across discounting dates.

Throughout this appendix, we specialize to the quadratic loss $L(x) = x^2$ and set $w_i = 1$, so that the population criterion reduces to $Q_H(\theta) = \mathbb{E}[(\bar{\epsilon}_i^{(H)}(\theta))^2]$.

To keep these objects distinct, we proceed in four steps. First, we define the horizon-specific population criterion and its pseudo-true minimizer. Second, we isolate the exact source of the population moment bias at θ_0. Third, we connect that moment bias to the shift in the population criterion's minimizer. Fourth, we present an exact MSE decomposition and then interpret the simulation results through a second-order finite-sample lens.

A.1 Setup and horizon-specific population target

For a given maximum compounding horizon H, let

$$\mathcal{T}_{i,H} \;:=\; \left\{t_i^0, t_i^0 + 1, \ldots, \min(t_i^0 + 12H, T_i^{\max})\right\},$$

where $T_i^{\max}$ denotes the largest discounting date available for fund i. Define the horizon-averaged pricing error by

$$\bar{\epsilon}_i^{(H)}(\theta) \;:=\; \frac{1}{|\mathcal{T}_{i,H}|} \sum_{\tau \in \mathcal{T}_{i,H}} \epsilon_{\tau,i}(\theta). \tag{27}$$

The associated population criterion is

$$Q_H(\theta) \;:=\; \mathbb{E}\Big[\big(\bar{\epsilon}_i^{(H)}(\theta)\big)^2\Big], \tag{28}$$

and the horizon-specific pseudo-true parameter is defined as its population minimizer:

$$\theta_H^\dagger \;:=\; \arg\min_\theta Q_H(\theta). \tag{29}$$

(Since this appendix works with $Q_H = \mathbb{E}[(\bar{\epsilon}_i^{(H)})^2]$ directly, $\theta_H^\dagger$ minimizes Q_H; in the main text, the sign-negated criterion $-\mathbb{E}[L(\cdot)]$ is maximized, but the first-order conditions are identical.)

The notation $\theta_H^\dagger$ emphasizes that this object is the target of the *population nonlinear least-squares criterion.* It is not defined as the zero of the averaged population moment, and those two objects need not coincide in general.

When $H = 0$ (equivalently, when the estimator uses only the benchmark NPV discounting date), the deal-level exact-pricing restriction (Equation (2)), whose unconditional counterpart $\mathbb{E}[\delta_{i,j}] = 0$ follows by iterated expectations, implies that the population criterion is centered at the structural parameter θ_0. Hence, under correct specification,

$$\theta_0^\dagger = \theta_0.$$

For $H > 0$, the inclusion of NFV terms generally perturbs the population criterion, so that

$$\theta_H^\dagger \neq \theta_0 \qquad \text{in general.}$$

It is convenient to define the horizon-specific population moment bias at the structural parameter:

$$b_H \;:=\; \mathbb{E}\Big[\bar{\epsilon}_i^{(H)}(\theta_0)\Big]. \tag{30}$$

Here, the expectation integrates over SDF and pricing-error randomness, holding the fund structure (number of deals J_i, investment dates, and horizon set $\mathcal{T}_{i,H}$) fixed. For a representative fund, b_H is therefore understood as the conditional-on-structure population mean of $\bar{\epsilon}_i^{(H)}(\theta_0)$. The next subsection shows that b_H is driven exactly by the NFV component of the Timing Risk Premium.

A.2 Population moment bias at θ_0

Starting from the average pricing error (Equation (12)) and the deal-level decomposition (Equation (3)), we can write

$$\bar{\epsilon}_i^{(H)}(\theta_0) = \frac{1}{|\mathcal{T}_{i,H}|} \sum_{\tau \in \mathcal{T}_{i,H}} \epsilon_{\tau,i}(\theta_0) = \frac{1}{|\mathcal{T}_{i,H}|} \sum_{\tau \in \mathcal{T}_{i,H}} \sum_{j=1}^{J_i} \frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}\, \delta_{i,j}. \tag{31}$$

Taking the unconditional expectation yields

$$b_H = \frac{1}{|\mathcal{T}_{i,H}|} \sum_{\tau \in \mathcal{T}_{i,H}} \sum_{j=1}^{J_i} \mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau} \delta_{i,j} \right]. \tag{32}$$

Because we now compute the unconditional expectation rather than conditioning on the deal investment date $\mathcal{F}_{t_{i,j}^{\text{Inv}}}$, we can decompose each (τ, j) term via the standard covariance identity $\mathbb{E}[XY] = \text{Cov}(X, Y) + \mathbb{E}[X]\mathbb{E}[Y]$:

$$\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau} \delta_{i,j} \right] = \underbrace{\text{Cov}\left(\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}, \delta_{i,j} \right)}_{\text{TRP}_{\tau,j}} + \underbrace{\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau} \right]}_{>0} \cdot \underbrace{\mathbb{E}[\delta_{i,j}]}_{=0}. \tag{33}$$

Under correct specification, $\mathbb{E}[\delta_{i,j}] = 0$, so only the covariance term remains. Note that the covariance in (33) is unconditional, whereas Lemma 1 works with conditional moments. The link is standard: the law of total covariance gives $\text{Cov}(X, Y) = \mathbb{E}[\text{Cov}(X, Y \mid \mathcal{F})] + \text{Cov}(\mathbb{E}[X \mid \mathcal{F}], \mathbb{E}[Y \mid \mathcal{F}])$; when iterated through $\mathcal{F}_{t_{i,j}^{\text{Inv}}}$, the second term involves the (zero) conditional mean of $\delta_{i,j}$. The key observation is that this unconditional covariance vanishes when $\tau \leq t_{i,j}^{\text{Inv}}$, because iterated expectations through $\mathcal{F}_{t_{i,j}^{\text{Inv}}}$ kill it, as in part (1) of Lemma 1. When $\tau \leq t_{i,j}^{\text{Inv}}$, the SDF ratio $\Psi_{t_{i,j}^{\text{Inv}}}/\Psi_\tau$ is determined entirely by realized factors up to time $t_{i,j}^{\text{Inv}}$, making it $\mathcal{F}_{t_{i,j}^{\text{Inv}}}$-measurable and allowing us to pass it outside the inner conditional expectation. Iterated expectations through $\mathcal{F}_{t_{i,j}^{\text{Inv}}}$ therefore give

$$\text{Cov}\left(\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}, \delta_{i,j} \right) = \mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau} \underbrace{\mathbb{E}\left[\delta_{i,j} \mid \mathcal{F}_{t_{i,j}^{\text{Inv}}} \right]}_{=0 \text{ by Equation (2)}} \right] = 0. \tag{34}$$

Hence only NFV terms contribute to b_H.

Lemma 2 (Exact source of the population moment bias)**.** *Under correct specification,*

$$b_H = \frac{1}{|\mathcal{T}_{i,H}|} \sum_{\tau \in \mathcal{T}_{i,H}} \sum_{j=1}^{J_i} \mathbf{1}\{\tau > t_{i,j}^{\text{Inv}}\} \, \text{Cov}\left(\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}, \delta_{i,j} \right). \tag{35}$$

In particular, the NPV terms contribute zero population bias, and the horizon-specific population moment bias is generated entirely by NFV terms through the Timing Risk Premium.

Proof. The result follows directly from (32), the covariance decomposition in (33), the condition $\mathbb{E}[\delta_{i,j}] = 0$, and the iterated-expectations argument ((34)) for all pairs (τ, j) satisfying $\tau \leq t_{i,j}^{\text{Inv}}$. □

Misspecification. Under model misspecification, $\mathbb{E}[\delta_{i,j}] \neq 0$ and the second term in (33) no longer vanishes. In that case, the averaged population moment contains both: (i) the TRP covariance term, and (ii) an additional mean-shift term $\mathbb{E}[\Psi_{t_{i,j}^{\mathrm{Inv}}}/\Psi_\tau] \cdot \mathbb{E}[\delta_{i,j}]$. Since this non-zero baseline permeates the NFV terms $\tau > t_{i,j}^{\mathrm{Inv}}$, it does not wash out with horizon averaging. Instead, the persistent misspecification bias bounds the average pricing error firmly away from zero, regardless of how large the compounding horizon H gets. Thus, under misspecification, $\mathcal{T}$-averaging no longer trades off only a small TRP-induced target shift against variance reduction; it also propagates a persistent baseline pricing-error component.

A.3 From population moment bias to target shift

Lemma 2 characterizes the average pricing moment at θ_0. To connect this to the parameter target, recall that the pseudo-true parameter $\theta_H^\dagger$ is defined by the first-order condition

$$\nabla_\theta Q_H(\theta_H^\dagger) = 0. \tag{36}$$

Assume that Q_H is twice continuously differentiable in a neighborhood of θ_0 and that its Hessian is nonsingular there. A first-order Taylor expansion of $\nabla_\theta Q_H(\theta)$ around θ_0 gives

$$0 = \nabla_\theta Q_H(\theta_0) + \nabla_\theta^2 Q_H(\bar{\theta}_H)\,(\theta_H^\dagger - \theta_0), \tag{37}$$

for some intermediate point $\bar{\theta}_H$ between θ_0 and $\theta_H^\dagger$. Rearranging yields

$$\theta_H^\dagger - \theta_0 = -\left[\nabla_\theta^2 Q_H(\bar{\theta}_H)\right]^{-1} \nabla_\theta Q_H(\theta_0). \tag{38}$$

Since

$$\nabla_\theta Q_H(\theta_0) = 2\,\mathbb{E}\left[\bar{\epsilon}_i^{(H)}(\theta_0)\,\nabla_\theta \bar{\epsilon}_i^{(H)}(\theta_0)\right], \tag{39}$$

the target shift is governed by the interaction between: (i) the horizon-specific population moment bias b_H, and (ii) the sensitivity of the averaged pricing error to θ.

Equation (38) makes precise the sense in which the TRP affects the population target: the covariance terms identified in Lemma 2 perturb the gradient of the population criterion at θ_0, thereby moving the minimizer from θ_0 to $\theta_H^\dagger$.

Remark 10 (Magnitude of the target shift)**.** *For $\tau > t_{i,j}^{\mathrm{Inv}}$, the covariance term in* (35) *satisfies the generic bound*

$$\left|\mathrm{Cov}\left(\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau},\, \delta_{i,j}\right)\right| \leq \sigma\left(\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}\right)\sigma(\delta_{i,j}),$$

by Cauchy–Schwarz. In Hansen–Jagannathan style, the variability of the SDF ratio is disciplined by the volatility of admissible pricing kernels over the corresponding horizon. Moreover, the averaging operator contributes the factor $1/|\mathcal{T}_{i,H}|$. Thus, although the target shift need not be zero for $H > 0$, its magnitude is economically constrained when the TRP covariances are moderate.

A.4 Exact MSE decomposition

We now separate the horizon-specific target shift from sampling error around that target. For any H, the estimator satisfies the exact identity

$$\hat{\theta}_H - \theta_0 = (\hat{\theta}_H - \theta_H^\dagger) + (\theta_H^\dagger - \theta_0). \tag{40}$$

Taking squared norms and expectations gives the following decomposition.

Proposition 6 (Exact MSE decomposition around the structural parameter)**.** *For any horizon H,*

$$\begin{aligned}\mathbb{E}\Big[\|\hat{\theta}_H - \theta_0\|^2\Big] &= \|\theta_H^\dagger - \theta_0\|^2 + 2(\theta_H^\dagger - \theta_0)^\top \mathbb{E}\Big[\hat{\theta}_H - \theta_H^\dagger\Big] \\ &\quad + \Big\|\mathbb{E}\Big[\hat{\theta}_H - \theta_H^\dagger\Big]\Big\|^2 + \operatorname{tr}\Big(\operatorname{Var}(\hat{\theta}_H)\Big).\end{aligned} \tag{41}$$

Proof. Write

$$\hat{\theta}_H - \theta_0 = (\hat{\theta}_H - \theta_H^\dagger) + (\theta_H^\dagger - \theta_0),$$

expand the squared norm, and then use

$$\mathbb{E}\Big[\|\hat{\theta}_H - \theta_H^\dagger\|^2\Big] = \Big\|\mathbb{E}[\hat{\theta}_H - \theta_H^\dagger]\Big\|^2 + \operatorname{tr}\Big(\operatorname{Var}(\hat{\theta}_H)\Big). \qquad \square$$

Proposition 6 isolates four conceptually distinct objects:

1. the *population target shift* $\|\theta_H^\dagger - \theta_0\|^2$,
2. the *cross-term* linking target shift and sampling bias,
3. the *finite-sample bias around the pseudo-true parameter* $\|\mathbb{E}[\hat{\theta}_H - \theta_H^\dagger]\|^2$,
4. the *sampling variance* $\operatorname{tr}(\operatorname{Var}(\hat{\theta}_H))$.

This decomposition is exact. What remains heuristic is the way these terms move with H, to which we now turn.

Heuristic link to second-order finite-sample bias. For nonlinear extremum estimators, standard second-order expansions imply that

$$\mathbb{E}[\hat{\theta}_H - \theta_H^\dagger] = O(n^{-1}),$$

with a constant depending on higher-order moments of the score and curvature; see, e.g., Nagar (1959); Rilstone et al. (1996); Newey and Smith (2004); Bao and Ullah (2007). In the present setting, those higher-order terms are naturally amplified when the objective contributions are noisy. This suggests the following heuristic channel: reducing the volatility of $\bar{\epsilon}_i^{(H)}(\theta)$ tends to reduce the second-order finite-sample bias around $\theta_H^\dagger$.

A convenient way to express the volatility of the horizon-averaged pricing error is

$$\operatorname{Var}\Big(\bar{\epsilon}_i^{(H)}(\theta)\Big) = a_{i,H}^\top \Sigma_{i,H}(\theta)\, a_{i,H}, \tag{42}$$

where $a_{i,H}$ is the $|\mathcal{T}_{i,H}| \times 1$ averaging vector with entries $1/|\mathcal{T}_{i,H}|$, and $\Sigma_{i,H}(\theta)$ is the covariance matrix of the vector

$$\big(\epsilon_{\tau,i}(\theta)\big)_{\tau \in \mathcal{T}_{i,H}}.$$

When the pricing errors at different discounting dates are positively but imperfectly correlated, averaging across τ reduces this variance relative to relying on a single date. This is the basic sense in which $\mathcal{T}$-averaging stabilizes the criterion in finite samples.

A.5 Interpretation and reconciliation with the simulation study

The simulation evidence in Section 4.5 can now be interpreted through Proposition 6.

At short horizons, the estimator relies on a small number of highly volatile pricing moments. This makes the nonlinear criterion unstable and can generate substantial finite-sample bias around the benchmark target. As H increases, averaging combines additional pricing moments that are correlated, but not perfectly so, thereby smoothing the objective and reducing finite-sample estimation error.

Against this benefit stands the population target shift documented in Lemma 2: once H is large enough to include NFV terms, the criterion is no longer centered at θ_0. The resulting TRP-induced shift is economically meaningful but, under the baseline calibration, remains modest relative to the initial finite-sample distortion at very short horizons.

This yields the central interpretation of the simulation results: for small to intermediate horizons, the stabilization benefit of $\mathcal{T}$-averaging dominates the incremental TRP cost; for sufficiently long horizons, the marginal gain from additional averaging weakens while the accumulation of NFV terms increases the scope for target shift. The empirically optimal horizon therefore need not be $H = 0$ and need not be arbitrarily large. Instead, the optimal horizon is *interior* and *DGP-dependent*. The sweet spot occurs roughly around the typical fund lifetime, at which point the variance-reduction benefit offsets the target shift induced by the Timing Risk Premium.

Remark 11 (Why we do not impose a global monotonicity theorem)**.** *We do not claim that each MSE component is globally monotone in H without additional restrictions. In particular, the aggregate TRP contribution may contain offsetting covariance terms, and the effect of averaging on estimator variance depends on the covariance structure of the pricing moments and on the local geometry of the nonlinear criterion. For this reason, the appendix establishes the exact decomposition and the source of the target shift, while the existence and location of the empirically optimal horizon are documented in the simulation study rather than asserted as a universal theorem.*

Remark 12 (Connection to the GMM and misspecification literatures)**.** *The role of $\mathcal{T}$-averaging is closely related to two familiar ideas. First, from a GMM perspective, averaging combines multiple imperfectly correlated pricing moments, which*

can improve finite-sample stability when those moments contain useful information. Second, from the perspective of estimation under misspecified moments, NFV terms can be viewed as informative but weakly biased components: they perturb the population target, yet may still reduce overall MSE when the induced target shift is small relative to the finite-sample stabilization they provide.

Summary. The main message is therefore not that the TRP "washes out" as H grows. Rather, $\mathcal{T}$-averaging works by trading off a controlled population target shift against a potentially large reduction in finite-sample estimation error. Under the baseline DGP, the latter effect dominates over an economically relevant range of horizons, which is why the simulation study favors horizons near the fund lifetime rather than the NPV-only benchmark.

A.6 Interpretation of NFV-based alpha

This subsection provides an auxiliary interpretation of NFV-based pricing errors for ex-post benchmarking. Unlike the baseline exact-pricing restriction in Equation (2), the decomposition below introduces an abnormal-performance component and is therefore used only as an interpretive device, not to define the structural estimation target.

Remark 13 (Measured Alpha vs. True Skill)**.** *The decomposition follows from the conditional covariance identity applied as in Lemma 1(3), writing* $\delta_{i,j} = a_{i,j}^{Skill} + (\delta_{i,j} - a_{i,j}^{Skill})$*, where* $a_{i,j}^{Skill} := \mathbb{E}[\delta_{i,j} \mid \mathcal{F}_{t_{i,j}^{\text{Inv}}}]$ *is the deal's abnormal return at inception. Specifically, we can decompose the expected NFV pricing error into two distinct components:*

$$\underbrace{\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Future}}}}{\Psi_\tau}\delta_{i,j} \mid \mathcal{F}_\tau\right]}_{\textit{Measured Alpha}} = \underbrace{\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\frac{a_{i,j}^{Skill}}{R_{f,t_{i,j}^{\text{Inv}}\to t_{i,j}^{\text{Future}}}} \mid \mathcal{F}_\tau\right]}_{\textit{True Skill}} + \underbrace{\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\text{Cov}\left(\frac{\Psi_{t_{i,j}^{\text{Future}}}}{\Psi_{t_{i,j}^{\text{Inv}}}}, \delta_{i,j} \mid \mathcal{F}_{t_{i,j}^{\text{Inv}}}\right) \mid \mathcal{F}_\tau\right]}_{\textit{Timing Risk Premium}} \tag{43}$$

where R_f *is the gross risk-free return over the holding period and* $a_{i,j}^{Skill}$ *is the deal-level abnormal return, distinct from the SDF intercept* α*. Even if a manager has no skill* ($a_{i,j}^{Skill} = 0$)*, the measured alpha will include the Timing Risk Premium.*

B Comparison to Driessen et al. (2012)

B.1 Comparison to simulation results

(Driessen et al., 2012, Table 1) report Monte Carlo results for $n = 50$ funds per vintage-year portfolio with an annualized idiosyncratic volatility of 108% ($\sigma_{\text{monthly}} = 0.3$). In our base case (S1), we use only $n = 20$ funds per vintage, which is more

realistic for older vintages and smaller private capital asset classes beyond Buyout and Venture Capital. Scenarios S2–S10 in Table 10 systematically vary the simulation design to identify which features drive the remaining discrepancy between our results and the Driessen et al. (2012) benchmark.

Several structural differences between the two simulation designs should be noted. First, Driessen et al. (2012) use a shifted lognormal distribution for the idiosyncratic error term (cf. their online appendix, Equation 9), which ensures that the error addend has a finite lower bound (approximately -1) and avoids the nonlinear distortions that arise when normally distributed errors occasionally produce extreme negative total returns. Second, they employ a "full Monte Carlo" approach and simulate the total market return R_{MKT} from a shifted lognormal distribution calibrated to S&P 500 moments (1980–2003), rather than conditioning on the historical path. Third, their deal investment timing follows a deterministic schedule (three deals per year, evenly spaced across the five-year investment period), rather than random uniform draws.

Interpretation of results. Under our original design with normal errors and $\sigma = 30\%$ (S3), the market beta is severely attenuated ($\hat{\beta} = 0.62$) and the spurious alpha (0.68%/month) is an order of magnitude larger than the benchmark (0.05%). Switching to the shifted lognormal error distribution (S4) modestly improves the beta estimate to 0.71 and reduces the alpha bias to 0.57%. Adding simulated market returns (S5) yields no further improvement ($\hat{\beta} = 0.66$), and deterministic investment timing (S7) produces a similarly attenuated beta of 0.70 without substantially reducing alpha variance.

The most effective modification is shortening the maximum deal holding period from 10 years to 5 years (S8), which significantly improves the beta estimate ($\hat{\beta} = 0.91$) and reduces the alpha bias to 0.30% without requiring lower idiosyncratic volatility. Combining this holding-period restriction with deterministic investment timing (S9) yields a similar beta estimate ($\hat{\beta} = 0.88$) and alpha bias (0.32%), with the deterministic timing contributing only a modest additional effect. By limiting the compounding duration, both scenarios restrict the geometric growth of idiosyncratic errors, reducing the frequency of degenerate alpha estimates and partially restoring the accuracy of the market beta. Tightening the alpha upper bound $\bar{\alpha}$ also dramatically affects results: at $\bar{\alpha} = 1\%$ (S6), 32% of draws hit the bound, and beta is close to the benchmark ($\hat{\beta} = 0.87$), but at the cost of artificially constraining the alpha search space. Combining all modifications—holding-period restrictions, deterministic timing, and a tighter alpha bound of $\bar{\alpha} = 1\%$ (S10)—produces our closest approximation to the benchmark. Under this configuration, the beta estimate improves to 0.93 and the alpha bias falls to 0.27%, though 10% of draws still hit the alpha bound. However, even under this most favorable scenario (S10), our estimator still exhibits substantially more beta attenuation ($\hat{\beta} = 0.93$ vs. 0.98) and alpha bias (0.27% vs. 0.05%) than the Driessen et al. (2012) benchmark. This sensitivity reveals a fundamental tension in the two-factor model: the alpha term absorbs residual nonlinearity that the linear SDF cannot capture, and bounding it arbitrarily distorts the beta estimate.

Overall, the large remaining discrepancy between our simulation results and those of Driessen et al. (2012), even after adopting their shifted lognormal errors

and simulated market returns, suggests that the favorable small-sample properties reported in Driessen et al. (2012) may be overly optimistic when the alpha upper bound is set more liberally ($\bar{\alpha}$ = 2%/month in our default configuration vs. $\bar{\alpha}$ = 1%/month in Driessen et al. (2012)). Across all scenarios with $\sigma = 30\%$, the beta estimates range from 0.62 to 0.93, well below the Driessen et al. (2012) benchmark of 0.98, confirming the estimator's sensitivity to the interplay between idiosyncratic noise, compounding Horizon, and alpha bounds.

Table 10: Comparison to simulation (Driessen et al., 2012, Table 1)

	Market Beta (β)					Alpha (α, monthly %)					
Scenario	True	Mean	SD	25%	75%	True	Mean	SD	25%	75%	% Bound
(S1) $n = 20$, $\sigma = 20\%$, Normal	1.00	0.84	0.40	0.61	1.05	0.00	0.22	0.28	0.04	0.32	0%
(S2) $n = 50$, $\sigma = 20\%$, Normal	1.00	0.89	0.29	0.74	1.03	0.00	0.13	0.16	0.03	0.19	0%
(S3) $n = 50$, $\sigma = 30\%$, Normal	1.00	0.62	0.72	0.12	1.03	0.00	0.68	0.70	0.13	1.08	0%
(S4) $n = 50$, $\sigma = 30\%$, Shifted LN	1.00	0.71	0.72	0.23	1.12	0.00	0.57	0.67	0.07	0.91	0%
(S5) $n = 50$, $\sigma = 30\%$, Shifted LN, Sim. MKT	1.00	0.66	0.79	0.11	1.06	0.00	0.71	0.79	0.03	1.34	0%
(S6) $n = 50$, $\sigma = 30\%$, Shifted LN, Sim. MKT, $\bar{\alpha} = 1\%$	1.00	0.87	0.69	0.41	1.24	0.00	0.48	0.50	0.06	1.00	32%
(S7) $n = 50$, $\sigma = 30\%$, Shifted LN, Sim. MKT, Det. Timing	1.00	0.70	0.79	0.19	1.12	0.00	0.65	0.80	-0.01	1.20	0%
(S8) $n = 50$, $\sigma = 30\%$, Shifted LN, Sim. MKT, Max. Hold 5	1.00	0.91	0.54	0.64	1.19	0.00	0.30	0.52	-0.03	0.47	0%
(S9) $n = 50$, $\sigma = 30\%$, Shifted LN, Sim. MKT, Max. Hold 5, Det. Timing	1.00	0.88	0.54	0.61	1.16	0.00	0.32	0.51	-0.01	0.53	0%
(S10) $n = 50$, $\sigma = 30\%$, Shifted LN, Sim. MKT, Max. Hold 5, Det. Timing, $\bar{\alpha} = 1\%$	1.00	0.93	0.50	0.65	1.17	0.00	0.27	0.39	-0.01	0.52	10%
Driessen et al. 2012, Table 1											
Benchmark Estimates ($n = 50$, $\sigma_{\text{monthly}} = 30\%$)	1.00	0.98	0.32	0.80	1.15	0.00	0.05	0.2	-0.06	0.15	n/a

Notes: This table reports the small-sample properties of the generalized estimator for Horizon 0 (i.e., treating funds as held to maturity), mimicking the structure of Table 1 in Driessen et al. (2012). We simulate 1,000 Monte Carlo draws. All scenarios use 20 vintage years (1986–2005) with 15 deals per fund; unless stated otherwise, $n = 20$ funds per vintage, $\sigma = 0.2$ monthly idiosyncratic volatility, and a normal error distribution. The simulation dimensions vary as follows: n = number of funds per vintage-year portfolio; σ = monthly idiosyncratic standard deviation; *Normal* = Gaussian error term (default); *Shifted LN* = shifted lognormal error term (cf. Driessen et al. (2012)); *Sim. MKT* = total market return drawn from a shifted lognormal (S&P 500 1980–2003 moments) with a constant risk-free rate of 4% annualized, instead of conditioning on the historical path; *Det. Timing* = deterministic (evenly spaced) deal investment dates instead of random; *Max. Hold 5* = maximum holding period of a deal limited to 5 years (default 10 years); $\bar{\alpha}$ = upper bound on the alpha parameter in the optimization (default $\bar{\alpha}$ = 2%/month). The *% Bound* column reports the fraction of draws where the alpha estimate reached the upper optimization bound $\bar{\alpha}$. The final row presents the Benchmark Estimates obtained directly from Table 1 in Driessen et al. (2012) for comparative purposes. Alpha estimates are monthly and presented in percentages.

B.2 Comparison to empirical results

In this section, we evaluate the empirical stability of single- and multi-factor SDF specifications by estimating these models on our dataset (for the US-Dollar North America subsample with maximum vintage 2010) and comparing the results directly with the benchmark estimates published by Driessen et al. (2012). A key distinction is the underlying private market dataset: we use the Preqin cash flow database, which is also employed by Korteweg and Nagel (2016) and Ang et al. (2018), whereas Driessen et al. (2012) rely on the Thomson Venture Economics (TVE) fund-level dataset, which suffers from well-documented data quality issues such as forward-filling of stale KPIs (Harris et al., 2014). (Driessen et al., 2012, Table 5) report results for Buyout (BO) and Venture Capital (VC) funds using both a two-factor model (S&P 500 MKT + α) and the Fama and French (1993) four-factor model

(MKT, SMB, HML, and α). In their two-factor specification, VC exhibits a market beta of 2.73 and a highly negative annualized alpha of -12.3%, while BO yields a market beta of 1.31 and an annualized alpha of -4.8%. These strongly negative alpha estimates can be at least partially attributed to the aforementioned data errors in TVE. Indeed, the BO alpha coefficient is statistically insignificant once bootstrap standard errors are taken into account, whereas the VC alpha remains significant. In their four-factor specification, the factor loadings are broadly plausible: BO exhibits $\hat{\beta}_{\text{MKT}} = 0.72$, $\hat{\beta}_{\text{SMB}} = 0.50$, and $\hat{\beta}_{\text{HML}} = 0.19$, while VC shows $\hat{\beta}_{\text{MKT}} = 2.20$, $\hat{\beta}_{\text{SMB}} = 1.83$, and $\hat{\beta}_{\text{HML}} = -0.48$. However, Driessen et al. (2012) note that the four-factor model does not significantly improve upon the two-factor specification and that the SMB and HML loadings are largely insignificant.

Table 11 and Table 12 present our empirical results for the single-factor (MKT), two-factor (α, MKT), three-factor (MKT, SMB, HML), and four-factor (α, MKT, SMB, HML) Fama and French (1993) models for BO, VC, and PE at Horizon 0, using the conservatively restricted sample with a maximum vintage-year cutoff of 2010.

We begin with the single-factor MKT model to establish a reliable baseline. Under this specification, market betas are consistently positive and theoretically plausible: the cross-validation fold averages ($\hat{\beta}_{\text{MKT}}$) are 1.23 for BO, 1.78 for VC, and 1.21 for PE under equal weighting (EW). The CV standard deviations verify that these estimates are highly stable across data subsets (e.g., CV standard deviations of 0.12 for BO and 0.09 for PE under EW). This result provides solid empirical backing that private equity returns exhibit strong, persistent co-movement with the broader market.

However, when explicitly allowing for an abnormal performance α intercept alongside the market factor, the estimation immediately collapses. In the two-factor model (α, MKT), the alpha parameter becomes highly sensitive to random, fund-specific noise compounded over a fund's lifetime. Even when restricting the sample to mature vintages, private equity datasets contain relatively few funds per year. As a result, there are simply not enough cross-sectional observations to adequately average out this idiosyncratic noise. Consequently, the optimizer overwhelmingly assigns inflated, positive alpha premiums (often hitting or nearing the 2% monthly bound, e.g., $\hat{\alpha} = 0.020$ for VC under EW and 0.008 for BO under FW), while the market betas are artificially driven toward zero (e.g., $\hat{\beta}_{\text{MKT}} = 0.14$ for BO and 0.18 for PE under FW). While asymptotic standard errors correspondingly explode or round to zero, the CV standard deviations reveal the actual parameter variation across data subsets. For instance, the cross-validation fold dispersion for the MKT estimate exceeds 0.6 for both BO and PE under FW, confirming that the collapsed point estimates are highly sensitive to the sample and untrustworthy.

This pathological behavior extends directly to the multi-factor models. When estimating the three-factor model (MKT, SMB, HML) *without* an alpha term, the estimates roughly mirror the MKT-only model (e.g., BO $\hat{\beta}_{\text{MKT}} = 1.35$ under EW), and the CV analysis shows reasonable out-of-sample stability, even though the asymptotic standard errors are vast. The SMB loading is consistently positive for BO and PE, but predominantly negative or near zero for VC. Notably, this sign for VC generally *reverses* the pattern reported by Driessen et al. (2012), who find a strongly

positive VC SMB loading ($\hat{\beta}_{\text{SMB}} = 1.83$). The discrepancy likely reflects differences in the underlying datasets (Preqin vs. TVE) and sample periods, underscoring the severe fragility of multi-factor estimates in private equity.

But once the alpha term is reintroduced in the four-factor model (α, MKT, SMB, HML), the framework fails for the same reasons as the two-factor model. Because the alpha term absorbs the bulk of the remaining variance, the other factor loadings swing erratically: the asymptotic MKT coefficient for BO drops substantially (to 0.25 under FW), while SMB and HML loadings reach extreme values. The pervasive asymptotic standard errors of exactly (0.000) in these models often coincide with the estimates hitting the boundary constraints. The CV analysis exposes this reality: rather than being estimated with zero error, the parameters swing erratically across hold-out sets, with CV standard deviations for the factor loadings routinely exceeding 0.4, indicating that the multi-factor models are highly unreliable when alpha is included.

In conclusion, our empirical evaluation provides a strong, objective corroboration of the simulation-based warnings from Section 4.5: with the limited cross-section of only 20–40 vintage-year portfolios currently available in commercial private equity databases, any generalized SDF model attempting to simultaneously extract multiple pricing factors alongside an explicit alpha parameter is severely under-identified given the available cross-section. The inclusion of an unconstrained or loosely constrained abnormal performance term confounds the signal with cross-sectional noise, significantly destabilizing the factor model estimates. Therefore, until substantially more granular and independent cross-sectional data becomes cleanly available, we conclude that the single-factor market model remains the most robust and estimable specification for private equity SDF performance evaluation in our implementations.

C Proofs for the horizon-specific asymptotic results

This subsection supplies detailed proofs for Proposition 2, Proposition 3, and Proposition 4. For notational convenience, fix a horizon $H \geq 0$ throughout and write

$$Z_i(\theta) := L\Big(\bar{\epsilon}_i^{(H)}(\theta)\Big), \qquad Q_{n,H}(\theta) = -\frac{1}{n}\sum_{i=1}^{n} Z_i(\theta).$$

Whenever convenient, re-index the observations by vintage year and within-vintage slot. That is, for vintage year $v = 1, \ldots, V$ and slot $\ell = 1, \ldots, n_v$, let $Z_{v,\ell}(\theta)$ denote the loss contribution of the ℓth unit in vintage year v. By Assumption 3(2), $n_v \leq \bar{n} < \infty$ uniformly in v. Hence the sample can be embedded in the increasing strip

$$\Lambda_V := \{(v, \ell) : v = 1, \ldots, V,\ \ell = 1, \ldots, \bar{n}\},$$

where empty slots are padded with zeros. Because $\bar{n}$ is fixed while $V \to \infty$, this embedding preserves the order of the sample size, and Assumption 3(2) implies

$$V \leq n = \sum_{v=1}^{V} n_v \leq \bar{n}V, \qquad \text{so that} \qquad n \asymp V.$$

Proof of Proposition 2. Let

$$\bar{Q}_{n,H}(\theta) := -\frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\big[Z_i(\theta)\big].$$

It suffices to show that

$$\sup_{\theta\in\Theta}\big|Q_{n,H}(\theta)-\bar{Q}_{n,H}(\theta)\big| \xrightarrow{p} 0, \qquad \sup_{\theta\in\Theta}\big|\bar{Q}_{n,H}(\theta)-Q_H(\theta)\big| \to 0,$$

because then the triangle inequality yields the claim.

We first verify the stochastic part. By Assumption 5(1), the array $\{\bar{\epsilon}_i^{(H)}(\theta) : i = 1,\dots,n\}$ is uniformly NED on an α-mixing vintage-year process $\{\eta_v\}_{v\geq 1}$. Since $L(\cdot)$ is continuous and Lipschitz by Assumption 5(2), $Z_i(\theta) = L(\bar{\epsilon}_i^{(H)}(\theta))$ inherits the same NED structure, uniformly in $\theta \in \Theta$. Moreover, Assumption 5(3) gives a uniform L^p envelope for some $p > 2$, and Assumption 5(4) gives stochastic equicontinuity in θ. Re-indexing by (v, ℓ) therefore produces an array of random fields on the strip Λ_V satisfying the primitive conditions of Jenish and Prucha (2009, Theorem 6). Assumption 3(2)–(4) ensures that the field is observed over an increasing domain, not an infill domain: the width of the strip is fixed, dependence is indexed by the vintage-year distance, and bounded lifetimes prevent the effective dependence neighborhoods induced by overlapping cash-flow windows from expanding with n. Hence

$$\sup_{\theta\in\Theta}\left|\frac{1}{n}\sum_{i=1}^{n}\Big(Z_i(\theta)-\mathbb{E}[Z_i(\theta)]\Big)\right| \xrightarrow{p} 0.$$

Equivalently,

$$\sup_{\theta\in\Theta}\big|Q_{n,H}(\theta)-\bar{Q}_{n,H}(\theta)\big| \xrightarrow{p} 0.$$

It remains to pass from the sample-specific mean criterion $\bar{Q}_{n,H}(\theta)$ to the population target $Q_H(\theta)$ defined in Equation (16). By the ergodic homogeneity invoked immediately after Equation (16), the Cesàro average of the expectations converges to the generic-unit expectation as $V \to \infty$; because $n \asymp V$, the same holds under n-normalization. Therefore,

$$\sup_{\theta\in\Theta}\big|\bar{Q}_{n,H}(\theta)-Q_H(\theta)\big| \to 0.$$

Combining the last two displays proves

$$\sup_{\theta\in\Theta}|Q_{n,H}(\theta)-Q_H(\theta)| \xrightarrow{p} 0.$$

□

Proof of Proposition 3. We verify the hypotheses of the standard argmax theorem directly.

Step 1: continuity of the population criterion. For each unit i, the map $\theta \mapsto \bar{\epsilon}_i^{(H)}(\theta)$ is continuous on Θ because $\mathcal{T}_{i,H}$ is finite and the discount-factor ratios $\Psi_{\tau,t}(\theta)$ are continuous in θ. Since $L(\cdot)$ is continuous, $\theta \mapsto Z_i(\theta)$ is continuous as well.

Assumption 5(3) supplies an integrable envelope uniformly in i and θ, so dominated convergence implies that

$$Q_H(\theta) = -\mathbb{E}\big[Z_i(\theta)\big]$$

is continuous on Θ. Because Θ is compact by Assumption 4, Q_H attains its maximum on Θ; by Assumption 6, this maximizer is unique and equals $\theta_H^\dagger$.

Step 2: uniform convergence. Proposition 2 yields

$$\sup_{\theta\in\Theta} |Q_{n,H}(\theta) - Q_H(\theta)| \xrightarrow{p} 0.$$

Step 3: argmax identification. Because $\hat{\theta}_H$ maximizes $Q_{n,H}$ over Θ, for every $\theta \in \Theta$,

$$Q_{n,H}(\hat{\theta}_H) \geq Q_{n,H}(\theta).$$

In particular, with $\theta = \theta_H^\dagger$,

$$Q_{n,H}(\hat{\theta}_H) \geq Q_{n,H}(\theta_H^\dagger).$$

Subtracting and adding the population criterion gives

$$\begin{aligned} 0 &\leq Q_H(\theta_H^\dagger) - Q_H(\hat{\theta}_H) \\ &\leq \big|Q_H(\theta_H^\dagger) - Q_{n,H}(\theta_H^\dagger)\big| + \big|Q_{n,H}(\hat{\theta}_H) - Q_H(\hat{\theta}_H)\big| \\ &\leq 2\sup_{\theta\in\Theta} |Q_{n,H}(\theta) - Q_H(\theta)|, \end{aligned}$$

which converges to zero in probability. Hence

$$Q_H(\hat{\theta}_H) \xrightarrow{p} Q_H(\theta_H^\dagger).$$

Now let $\{n_k\}$ be any subsequence. By compactness of Θ, there exists a further subsequence, still denoted $\{n_k\}$ for notational simplicity, such that $\hat{\theta}_H \to \bar{\theta}$ almost surely along that subsequence for some random $\bar{\theta} \in \Theta$. By continuity of Q_H,

$$Q_H(\hat{\theta}_H) \to Q_H(\bar{\theta}) \qquad \text{almost surely along the subsequence.}$$

Since the same subsequence satisfies $Q_H(\hat{\theta}_H) \to Q_H(\theta_H^\dagger)$ in probability, it follows that $Q_H(\bar{\theta}) = Q_H(\theta_H^\dagger)$ almost surely. By uniqueness of the maximizer, $\bar{\theta} = \theta_H^\dagger$ almost surely. Therefore every convergent subsequence of $\hat{\theta}_H$ has the same limit $\theta_H^\dagger$, which proves

$$\hat{\theta}_H \xrightarrow{p} \theta_H^\dagger.$$

This is exactly the conclusion of Newey and McFadden (1994, Theorem 2.1) specialized to the present criterion. □

Proof of Proposition 4. Define the sample score and sample Hessian by

$$S_{n,H}(\theta) := \nabla_\theta Q_{n,H}(\theta) = -\frac{1}{n}\sum_{i=1}^{n} s_i^{(H)}(\theta), \qquad \mathcal{H}_{n,H}(\theta) := \nabla_{\theta\theta} Q_{n,H}(\theta).$$

Because $\theta_H^\dagger$ lies in the interior of Θ and $\hat{\theta}_H \xrightarrow{p} \theta_H^\dagger$ by Proposition 3, $\hat{\theta}_H$ is an interior maximizer with probability approaching one. Hence the first-order condition holds with probability approaching one:

$$S_{n,H}(\hat{\theta}_H) = 0.$$

Applying a mean-value expansion of $S_{n,H}(\cdot)$ around $\theta_H^\dagger$ yields

$$0 = S_{n,H}(\theta_H^\dagger) + \mathcal{H}_{n,H}(\tilde{\theta}_H)\big(\hat{\theta}_H - \theta_H^\dagger\big),$$

where $\tilde{\theta}_H$ lies on the line segment joining $\hat{\theta}_H$ and $\theta_H^\dagger$. Rearranging gives

$$\sqrt{n}\,\big(\hat{\theta}_H - \theta_H^\dagger\big) = -\mathcal{H}_{n,H}(\tilde{\theta}_H)^{-1}\sqrt{n}\,S_{n,H}(\theta_H^\dagger). \tag{44}$$

We now analyze the two factors on the right-hand side.

Step 1: asymptotic distribution of the score. By definition,

$$\sqrt{n}\,S_{n,H}(\theta_H^\dagger) = -\frac{1}{\sqrt{n}}\sum_{i=1}^{n} s_i^{(H)}(\theta_H^\dagger).$$

Assumption 4 of Proposition 4 imposes a central limit theorem for these score contributions. In the present application, the required weak dependence is generated by the vintage-year random field, and the maintained NED and mixing conditions are exactly those covered by Jenish and Prucha (2012, Theorem 4) under increasing-domain asymptotics. Therefore,

$$\sqrt{n}\,S_{n,H}(\theta_H^\dagger) \xrightarrow{d} \mathcal{N}(0, \Lambda_H),$$

where

$$\Lambda_H = \lim_{n\to\infty} n \cdot \mathrm{Var}\Big[\nabla_\theta Q_{n,H}(\theta_H^\dagger)\Big].$$

Step 2: convergence of the sample Hessian. Because $\hat{\theta}_H \xrightarrow{p} \theta_H^\dagger$, the intermediate point $\tilde{\theta}_H$ also satisfies $\tilde{\theta}_H \xrightarrow{p} \theta_H^\dagger$. Hence, with probability approaching one, $\tilde{\theta}_H \in \mathcal{N}_H$. Assumption 7 then gives

$$\mathcal{H}_{n,H}(\tilde{\theta}_H) - \nabla_{\theta\theta} Q_H(\tilde{\theta}_H) \xrightarrow{p} 0.$$

Since Q_H is twice continuously differentiable in a neighborhood of $\theta_H^\dagger$, continuity of its Hessian and $\tilde{\theta}_H \xrightarrow{p} \theta_H^\dagger$ imply

$$\nabla_{\theta\theta} Q_H(\tilde{\theta}_H) \xrightarrow{p} \nabla_{\theta\theta} Q_H(\theta_H^\dagger) = \mathcal{H}_H.$$

Therefore,

$$\mathcal{H}_{n,H}(\tilde{\theta}_H) \xrightarrow{p} \mathcal{H}_H.$$

By assumption, $\mathcal{H}_H$ is nonsingular, so the continuous mapping theorem yields

$$\mathcal{H}_{n,H}(\tilde{\theta}_H)^{-1} \xrightarrow{p} \mathcal{H}_H^{-1}.$$

Step 3: conclusion. Substituting the previous two steps into Equation (44) and applying Slutsky's theorem gives

$$\sqrt{n}\left(\hat{\theta}_H - \theta_H^\dagger\right) \xrightarrow{d} \mathcal{N}\big(0,\, \mathcal{H}_H^{-1}\Lambda_H\mathcal{H}_H^{-1}\big).$$

This is the extremum-estimator central limit theorem of Newey and McFadden (1994, Theorem 3.1), with the non-iid score CLT supplied by the increasing-domain NED random-field result of Jenish and Prucha (2012, Theorem 4). □

Table 11: Estimation Results: 1-, 2-, 3-, and 4-Factor Models for Buyout and Venture Capital

	Equal-Weighted (EW)				Fund-Weighted (FW)			
Model	α	MKT	SMB	HML	α	MKT	SMB	HML
Buyout (BO)								
1-Factor (Asymp.)		1.268				1.197		
		(1.162)				(1.177)		
1-Factor (CV)		1.234				1.142		
		(0.119)				(0.137)		
2-Factor (Asymp.)	0.004	1.005			0.008	0.140		
	(0.006)	(1.070)			(0.189)	(25.676)		
2-Factor (CV)	0.006	0.588			0.008	0.122		
	(0.004)	(0.481)			(0.005)	(0.718)		
3-Factor (Asymp.)		1.389	1.296	0.415		1.252	1.870	0.296
		(1.588)	(4.991)	(0.988)		(2.471)	(6.876)	(2.989)
3-Factor (CV)		1.354	1.144	0.391		1.198	1.544	0.196
		(0.177)	(0.382)	(0.114)		(0.148)	(0.510)	(0.274)
4-Factor (Asymp.)	0.004	1.005	0.006	0.008	0.007	0.251	0.147	0.115
	(0.003)	(0.000)	(0.000)	(0.000)	(0.009)	(0.000)	(0.000)	(0.000)
4-Factor (CV)	0.004	0.901	0.115	-0.005	0.007	0.420	0.548	-0.159
	(0.002)	(0.215)	(0.241)	(0.122)	(0.007)	(0.774)	(1.090)	(0.907)
Venture Capital (VC)								
1-Factor (Asymp.)		1.887				1.588		
		(1.768)				(1.655)		
1-Factor (CV)		1.775				1.592		
		(0.405)				(0.363)		
2-Factor (Asymp.)	0.020	0.467			-0.001	1.735		
	(0.121)	(19.892)			(0.050)	(11.554)		
2-Factor (CV)	0.014	0.732			0.004	1.409		
	(0.009)	(0.186)			(0.010)	(0.589)		
3-Factor (Asymp.)		2.013	-0.364	1.425		1.943	-0.444	1.395
		(0.755)	(2.207)	(0.440)		(0.856)	(1.689)	(1.521)
3-Factor (CV)		1.911	-0.330	1.209		1.887	0.051	1.112
		(0.453)	(0.430)	(0.946)		(0.666)	(0.720)	(1.444)
4-Factor (Asymp.)	0.020	1.384	0.680	4.096	0.001	1.620	-0.534	1.112
	(0.000)	(0.000)	(0.000)	(0.000)	(0.003)	(0.000)	(0.000)	(0.000)
4-Factor (CV)	0.012	1.090	-0.171	1.534	0.005	1.780	-0.053	1.437
	(0.008)	(0.373)	(0.336)	(1.511)	(0.010)	(0.558)	(0.722)	(1.941)

Note: This table presents the estimation results of the single-factor (MKT), 2-factor (α, MKT), Fama-French 3-factor (MKT, SMB, HML), and 4-factor (α, MKT, SMB, HML) models. Cash flows are discounted to horizon 0. Both asymptotic (Asymp.) full-sample estimates and cross-validation (CV) means across hold-out sets are reported. Standard errors (or standard deviations of the estimates for CV) are reported in parentheses below the estimates. The estimation uses Equal-Weighted (EW) and Fund-Size-Weighted (FW) cash flows. The upper bound for α was constrained to 0.02.

Table 12: Estimation Results: 1-, 2-, 3-, and 4-Factor Models for Private Equity

	Equal-Weighted (EW)				Fund-Weighted (FW)			
Model	α	MKT	SMB	HML	α	MKT	SMB	HML
Private Equity (PE)								
1-Factor (Asymp.)		1.216				1.134		
		(0.793)				(1.171)		
1-Factor (CV)		1.214				1.083		
		(0.087)				(0.136)		
2-Factor (Asymp.)	0.003	1.003			0.008	0.178		
	(0.001)	(0.801)			(0.595)	(88.860)		
2-Factor (CV)	0.004	0.912			0.007	0.154		
	(0.002)	(0.259)			(0.005)	(0.630)		
3-Factor (Asymp.)		1.252	0.790	0.471		1.177	1.695	0.240
		(13.846)	(4.351)	(10.009)		(4.248)	(4.996)	(2.624)
3-Factor (CV)		1.210	0.522	0.548		1.140	1.418	0.169
		(0.186)	(0.328)	(0.168)		(0.131)	(0.443)	(0.223)
4-Factor (Asymp.)	0.003	1.003	0.003	0.006	0.009	0.035	0.095	-0.177
	(0.001)	(0.000)	(0.000)	(0.000)	(0.008)	(0.000)	(0.000)	(0.000)
4-Factor (CV)	0.004	0.924	0.035	0.033	0.006	0.367	0.343	-0.127
	(0.002)	(0.226)	(0.091)	(0.075)	(0.005)	(0.460)	(1.075)	(0.836)

Note: This table presents the estimation results of the single-factor (MKT), 2-factor (α, MKT), Fama-French 3-factor (MKT, SMB, HML), and 4-factor (α, MKT, SMB, HML) models. Cash flows are discounted to horizon 0. Both asymptotic (Asymp.) full-sample estimates and cross-validation (CV) means across hold-out sets are reported. Standard errors (or standard deviations of the estimates for CV) are reported in parentheses below the estimates. The estimation uses Equal-Weighted (EW) and Fund-Size-Weighted (FW) cash flows. The upper bound for α was constrained to 0.02.